Patchworking Curves

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Abstract

Our aim is to study the number of connected components of degree-m curves in $\mathbb{R}^n$ and $\mathbb{RP}^n$. Along the way, we discuss about the projective plane and its properties, we study Harnack's inequality and Milnor-Thom theorem, and we introduce the patchworking technique as a way of constructing real algebraic plane curves. Finally, our main result is a lower bound on the number of components of degree-m surfaces in $\mathbb{R}^3$.

Declaration

I declare that this thesis was composed by myself and that the work contained therein is my own, except where explicitly stated otherwise in the text.

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Chapter 1

Introduction

In 1876, Harnack posited the question 'topological classification of non-singular curves of a given degree in $\mathbb{RP}^2$' ([5]).

The main objective of this paper is to study the number of connected components of smooth degree-m curves in $\mathbb{R}^n$, which we denote $C_{n,m}$.

Chapter 2 mainly introduces the background content we will need throughout the paper. Namely, we discuss the real projective plane $\mathbb{RP}^2$, real plane algebraic curves, and their key properties. We then proceed to discuss two important inequalities which help us bound from above our quantity of interest, $C_{n,m}$ - Harnack Inequality, which gives us an upper bound for the number of components in $\mathbb{RP}^2$:

$$C_{2,m} \leq \frac{(m-1)(m-2)}{2} + 1,$$

and Milnor-Thom's theorem which gives an upper bound on the number of components in $\mathbb{R}^n$

$$C_{n,m} \leq m(2m-1)^{n-1}.$$

The connection between real algebraic curves and combinatorial geometry is done via a technique introduced by O. Y. Viro ([7]) - patchworking which we introduced in 3. We then proceed to discuss numerous applications of patchworking: from constructing Harnack curves to deriving bounds on $C_{2,m}$. We also briefly mention the original application of the technqiue, a construction of a counterexample to the Ragsdale conjecture.

Finally, we discuss a three dimensional version of the patchworking theorem. Throughout the chapter, we work torwards deriving our main result, a lower bound on $C_{3,m}$.

Chapter 2

A study of Real Plane Curves

The work in this paper is built on an understanding of the fundamentals of projective plane. As such, with close reference to [4], [3], [2], and [14] (unless otherwise specified), we begin by introducing algebraic plane curves in the real plane and in the real projective plane.

We summarily illustrate the key differences between the two types of curves by use of examples.

By means of the Harnack inequality 2.2.1, we proceed to give an upper bound on the number of components. In a similar manner we discuss Milnor-Thom theorem 2.3.1 which gives us a bound on the number of components of curves in $\mathbb{R}^n$. Finally, we reiterate Ragsdale conjecture which makes a claim about the arrangement of curves in $\mathbb{RP}^2$.

2.1 Curves in $\mathbb{RP}^2$

Before attempting to describe the real projective plane, we offer some motivation for its use. As stated in [2], while two triples in $\mathbb{R}^3$ are said to be equal if and only if they are equal component-wise, it can often be helpful to have a weaker notion of equality. One such example (according to [7]) was Netwon's study [11] on degree-3 curves: while he found 99 classes of curves in $\mathbb{R}^2$, he also found that all of them can be obtained as plane sections of 5 cubic cones in $\mathbb{R}^3$. Going forward, we will discover that often we can derive interesting properties about curves in $\mathbb{R}^2$ starting from the properties of curves in $\mathbb{RP}^2$.

2.1.1 Real projective plane

We use [2] as our main source. We start by considering the following relation $\sim$ on $\mathbb{R}^3 \setminus (0,0,0)$:

$$(x, y, z) \sim (x', y', z') \Leftrightarrow (x, y, z) = (\lambda x', \lambda y', \lambda z') \text{ for some } \lambda \in \mathbb{R},\ \lambda \neq 0.$$

We can easily show that $\sim$ is an equivalence relation: reflexivity and symmetry follow directly from the reflexivity and symmetry of the equality relation. For transitivity, suppose $(x, y, z) \sim (x', y', z')$ and $(x', y', z') \sim (x'', y'', z'')$. So there exist non-zero $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $(x, y, z) = (\lambda_1 x', \lambda_1 y', \lambda_1 z')$ and $(x', y', z') = (\lambda_2 x'', \lambda_2 y'', \lambda_2 z'')$. Thus, $(x, y, z) = (\lambda_1\lambda_2 x'', \lambda_1\lambda_2 y'', \lambda_1\lambda_2 z'')$ and $\lambda_1\lambda_2 \in \mathbb{R}$, $\lambda_1\lambda_2 \neq 0$, i.e. $(x, y, z) \sim (x'', y'', z'')$.

Now, for every real triple $(x, y, z) \neq (0, 0, 0)$, let us denote by $[x : y : z]$ the set

$$[x : y : z] = \{(\lambda x, \lambda y, \lambda z) \mid \lambda \in \mathbb{R},\ \lambda \neq 0\}.$$

Thus, $[x : y : z]$ is the equivalence class of the point (x, y, z) with respect to the relation $\sim$ defined above

Definition 2.1.1. *We define the real projective plane $\mathbb{RP}^2$ to be the set of all equivalence classes $[x : y : z]$ where $x, y, z \in \mathbb{R}$ and $(x, y, z) \neq 0$, i.e.*

$$\mathbb{RP}^2 = (\mathbb{R}^3 \setminus (0, 0, 0))/\sim .$$

We use [13] as our source. We can think of the real projective plane as the space of lines passing through the origin in the real three dimensional space $\mathbb{R}^3$. A line through the origin is determined by its direction, so we can consider the space of 3-dimensional unit vectors starting at the origin (which is just the unit sphere centred at the origin) in which we identify a unit vector with its opposite (since a line has no notion of 'sense'). Thus, we can think of $\mathbb{RP}^2$ as the unit sphere modulo this 'opposite-identifying' map. Let us state this formally:

Definition 2.1.2. *Let $\sim'$ be the relation on the unit sphere S^2 given by $x \sim' -x$. We call $\sim'$ the antipodal map.*

So we can give an equivalent definition of $\mathbb{RP}^2$:

Proposition 2.1.3. $\mathbb{RP}^2 = S^2/\sim'$.

Remark 2.1.4. We can derive this from our first definition. We have $\mathbb{RP}^2 = (\mathbb{R}^3 \setminus (0, 0, 0))/\sim$. Let us do the quotient in two steps. First, let us slightly redefine $\sim$ so that

$$(x, y, z) \sim (x', y', z') \Leftrightarrow (x, y, z) = (\lambda x', \lambda y', \lambda z') \text{ for some } \lambda > 0.$$

Taking the quotient will give the unit sphere S^2. Now, we take the quotient by the antipodal map.

Now, since we identify points on the unit sphere using the antipodal map, it is enough to keep just one hemisphere (Figure 2.1).

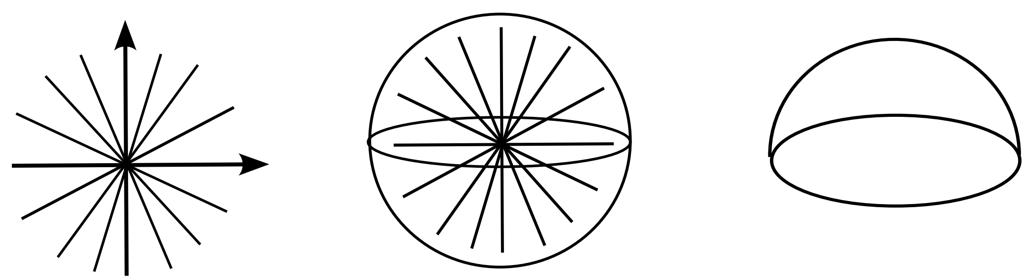


Figure 2.1: $\mathbb{RP}^2$ seen as 1. the space of lines passing through the origin in $\mathbb{R}^3$; 2. the unit sphere centred at the origin; 3. the the upper closed hemisphere. Source of the figure.

We keep the upper (closed) hemisphere, and we only need to be careful with the identification at the equator. The upper closed hemisphere is topologically equivalent to a disc (i.e. we can find a homeomorphism between the two), so we can 'flatten it out' to the plane of the equator: we obtain a disc in which we identify the opposite points on the boundary (Figure 2.2). We will often make use of this representation of $\mathbb{RP}^2$.

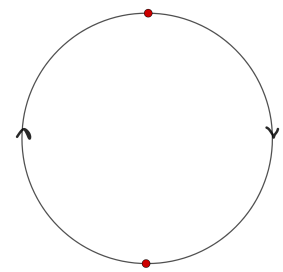


Figure 2.2: $\mathbb{RP}^2$ as a disc

Remark 2.1.5. A non-orientable surface is a surface which contains the Möbius strip. It is easy to observe that $\mathbb{RP}^2$ is non-orientable using the representation shown in 2.2.

[2] The real projective plane $\mathbb{RP}^2$ consists of $\mathbb{R}^2$ and a line at infinity, L_∞.

Let us consider the subset $U_z \subset \mathbb{RP}^2$ defined as following

$$U_z = \{[x : y : z] \mid x, y, z \in \mathbb{R}\,, z \neq 0\}$$

Since $z \neq 0$, we have $[x : y : z] = \left[\frac{x}{z} : \frac{y}{z} : 1\right]$. So every element of U_z is determind by two numbers, $\frac{x}{z}$ and $\frac{y}{z}$. Thus, we have a map $U_z \to \mathbb{R}^2$ given by

$$[x : y : z] \mapsto \left(\frac{x}{z}, \frac{y}{z}\right).$$

It is easy to observe that this map is a bijection. Thus, we can identify U_z and $\mathbb{R}^2$.

Now, let us consider $\mathbb{RP}^2 \setminus U_z$:

$$\mathbb{RP}^2 \setminus U_z = \{[x : y : 0] \mid x, y \in \mathbb{R} \text{ and at least one of them is non-zero}\}.$$

So we can identify $\mathbb{RP}^2 \setminus U_z$ with the real projective line, i.e. the line at infinity, L_∞.

Thus, we can view $\mathbb{RP}^2$ as being formed of two disjoint pieces: $\mathbb{R}^2$ and the line at infinity, L_∞.
Remark 2.1.6. We could have chosen either x or y to form subsets U_x or U_y.

We can define the real projective n-dimensional space in a similar fashion. If we let $\sim$ be a relation on $\mathbb{R}^{n+1}$ such that

$$x \sim x' \Leftrightarrow x = \lambda x' \text{ for some } \lambda \in \mathbb{R},$$

where x is a $n+1$-tuple, we can then define $\mathbb{RP}^n$ as $\mathbb{RP}^n = \mathbb{R}^{n+1} \setminus (\underbrace{0, 0, .., 0}_{n+1 \text{ times}})/\sim$.

Equivalently, if we let $\sim'$ be the antipodal map defined on the unit n-sphere S^n, i.e. $x \sim' -x$, then we can define the real projective n-dimensional space as $\mathbb{RP}^n = S^n/\sim'$.

2.1.2 Real Algebraic Projective Plane Curves

Now, let us introduce curves in $\mathbb{RP}^2$, i.e. real algebraic projective plane curves (or, simply, plane curves). We again use [2] as our main source. We note that this exposition is rather brief (at the cost of rigour), its main aim being to introduce the notions which will come in useful later.

First, let us recall the definition of an affine curve:
Definition 2.1.7. *An real affine algebraic plane curve of degree d, C_d, is a subset given by the equation*

$$g_d(x, y) = 0$$

where g_d is a real homogenous polynomial of degree d.

In contrast, when talking about projective curves, we want to consider the solutions on the line at infinity:
Definition 2.1.8. *A real plane projective curve of degree d, C_d, is a subset in $\mathbb{RP}^2$ given by an equation*

$$f_d(x, y, z) = 0$$

where f_d is a homoegenous real polynomial of degree d (with $d \in \mathbb{N}$).
Remark 2.1.9. Rather confusingly, some authors consider the curve C_d to be the polynomial f_d, and call the subset given by $f_d(x, y, z) = 0$ the subset of real points of C_d, denoted $\mathbb{R}C_d$. As a word of warning for further chapters, Itenberg and Viro ([7]) follow this convention, and so we will cite their results also using this convention.
Remark 2.1.10. Real plane projective curves of degree 1 are called lines and those of degree 2 are called conics.
Definition 2.1.11.

1. *Let C_d be a curve in $\mathbb{RP}^2$ given by the homogenous polynomial f_d, and let be $[a : b : c]$ be a point on C_d. Then we say $[a : b : c]$ is a singular point of C_d if $[a : b : c]$ is a solution to the system of equations:*

$$\frac{\partial f_d(x, y, z)}{\partial x} = 0 \qquad \frac{\partial f_d(x, y, z)}{\partial y} = 0 \qquad \frac{\partial f_d(x, y, z)}{\partial z} = 0. \tag{2.1}$$

2. *If the system 2.1 has no solutions in $\mathbb{RP}^2$, then all points of C_d are non-singular. In this case, we call the curve C_d non-singular.*

Example 2.1.12. Ellipses, hyperbolae, and parabolae have different shapes in $\mathbb{R}^2$, but the same shape in $\mathbb{RP}^2$.

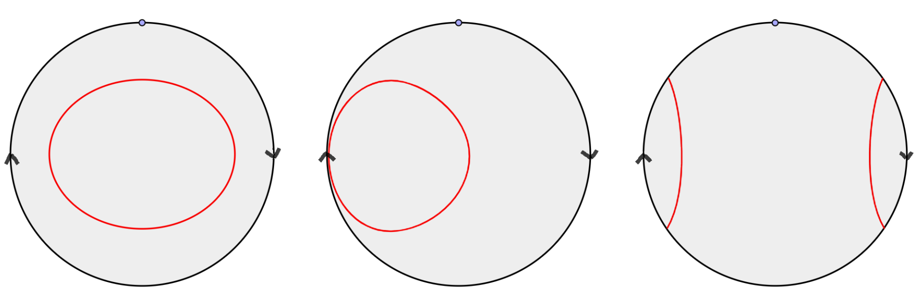


Figure 2.3: Ellipse, parabola and hyperbola in $\mathbb{RP}^2$. Source: ..

It is easy to observe that they do not have the same shape in $\mathbb{R}^2$: an ellipse is connected, while a hyperbola has two connected components tending to infinity, while a parabola has two branches tending to infinity.

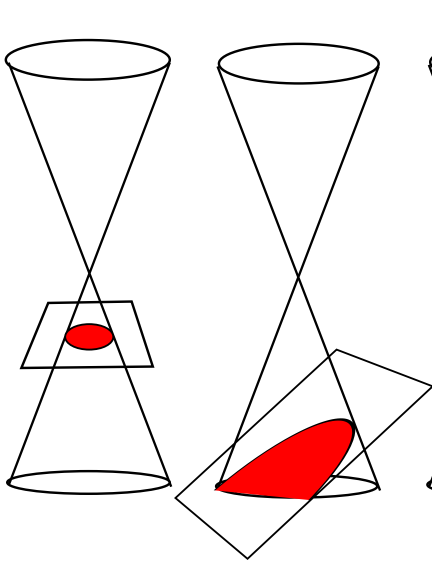


Figure 2.4: Circle, parabola, ellipse, and hyperbola in $\mathbb{RP}^2$. Reproduced from [1]

Example 2.1.13. All lines (i.e. real curves of the form $ax + by + c = 0$ with $a, b \in \mathbb{R}$, $a, b \neq 0$) have the same shape in $\mathbb{R}^2$, and thus in $\mathbb{RP}^2$.

Let us consider the line $x - y = 0$. Informally, one can obtain any other line by rotating and translating this line.

Example 2.1.14. Consider the family of curves $C_s = \{(x-s)(x-2)^2+y^2-1)((x+2)^2+y^2-1) = 0\}$ for $s \in \mathbb{R}$. Then, C_4 and C_{-4} have the same shape in $\mathbb{R}^2$, but a different shape than C_0. However, all three curves have the same shape in $\mathbb{RP}^2$.

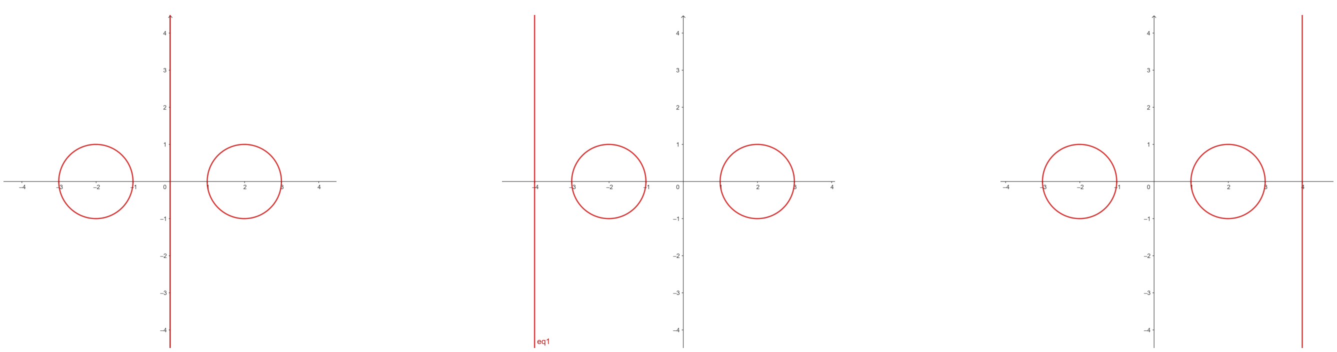


Figure 2.5: C_0, C_{-4}, and C_4 (from left to right)

Note that we can obtain C_{-4} by rotation starting from C_4, thus they have the same shape in $\mathbb{R}^2$. However, if we use the representantion of $\mathbb{R}P^2$ from 2.2, we can easily see that all three curves have the same shape.

Properties

Theorem 2.1.15 (Bézout's theorem)**.** *Let C_{d_1}, C_{d_2} be two curves in $\mathbb{RP}^2$ of degree d_1 respectively d_2 that have no common components. Then $|C_{d_1} \cap C_{d_2}| \leq d_1 d_2$.*

Remark 2.1.16. If the curves had a common component, then $|C_{d_1} \cap C_{d_2}|$ would be infinite.
Remark 2.1.17. If we count the points of intersection with multiplicity, then we can say that $|C_{d_1} \cap C_{d_2}| = d_1 d_2$. The notion of 'multiplicity' is intuitively the one we meet in algebra. For example, consider a tangent line at a curve, as shown in Figure 2.6. The point at which the line is tangent to the curve has multiplicity 2.

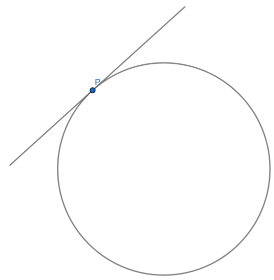


Figure 2.6: Tangent line at a curve

However, we omit here a formal definition of multiplicity, and we refer the reader to section 1.d of [2] for more details.
Example 2.1.18. Consider the degree-2 curves given by $x^2 + y^2 - 1 = 0$ and $\frac{1}{2}x^2 + 2y^2 - 1 = 0$ shown in Figure 2.7. They intersect in 4 points which is the maximum possible.

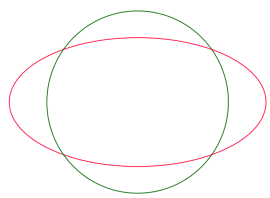


Figure 2.7: Intersection of two degree-2 curves

Example 2.1.19. Let L_1 be the line $x - y - 1 = 0$ and L_2 the line $x - y - 5 = 0$. Since we are in $\mathbb{RP}^2$, the two lines intersect at L_∞, the line at infinity.
Corollary 2.1.20. *Let L be a line and C_d a curve of degree d in $\mathbb{RP}^2$. Then $|L \cap C_d| \leq d$.*

2.2 The Harnack Inequality

The following inequality serves as a sharp upper bound on the number of components of a curve in $\mathbb{RP}^2$ of a given degree m. The proof in its entirety is sourced from [5].
Theorem 2.2.1 (The Harnack Inequality)**.** *Let $m \in \mathbb{N}$. Then a nonsingular real plane algebraic curve of degree of m in $\mathbb{RP}^2$ has at most*

$$\frac{(m-1)(m-2)}{2} + 1$$

connected components.

Proof. We omit the proof here, but we refer the reader to Section 11.6 of [8]. We just note that we can show this by contradiction: if we let A be a curve of degree m which has more than

$M = \frac{(m-1)(m-2)}{2} + 1$ connected components, we can show it is possible to construct a curve B of degree $m-2$ that passes through M points on M connected components of A and $m-3$ points on one more connected component. We would then have to show that A and B interesect in more than $m(m-2)$ points, which is not possible by the Bézout theorem.

□

Remark 2.2.2. We name the curves that achieve the maximal number of connected components M-curves. There are numerous algorithms to construct such curves.

Harnack's method involed the repeated perturbation of a circle and a line. His curves obeyed special properties: the depth of nested ovals cannot exceed 2, and the number of inner and outer ovals were given as quadratic functions of the degree of the curve. Figure 2.8 shows an M-curve of degree 6 obtained using Harnack's method. Another well-known construction of M-curves is

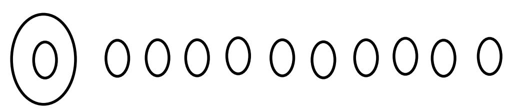


Figure 2.8: A Harnack curve of degree 6 (taken from [7]). The curve has 10 outer ovals and 1 inner oval

Hilbert's method which involves two ellipses which are perturbed by carefully chosen lines. We illustrate Hilbert's construction for an M-curve of degree 4 in Figure 2.9.

2.3 The Milnor-Thom Theorem

Milnor [10] and Thom [15] proved that the sum of the Betti numbers of the set of zeros of polynomials of degree $m > 0$ in $\mathbb{R}^n$ is at most $m(2m-1)^{n-1}$. In particular, since Betti numbers are non-negative and the first Betti number, B_0, is just the number of connected components of the curve given by the polynomial, we can state the following:

Theorem 2.3.1 (Milnor-Thom)**.** *Let $V \subset \mathbb{R}^n$ be a real algebraic hypersurface of degree $m > 0$ i.e.*

$$V = \left\{ \sum_{i_1,..,i_n \geq 0} c_{i_1,..,i_n} x_1^{i_1} \ldots x_n^{i_n} = 0 \right\} \subseteq \mathbb{R}^n.$$

Then the number of connected components of V is at most $m(2m-1)^{n-1}$.

We omit the proof here, but we refer the reader to [10].

Example 2.3.2. Recall that we denote by $C_{n,m}$ the maximal number of connected components of a degree m real hypersurface in $\mathbb{R}^n$. By Milnor-Thom, we have

$$C_{2,1} \leq 1$$

$$C_{2,2} \leq 6.$$

Indeed, we have that $C_{2,1} = 1$ (a line can only have one component) and $C_{2,2} = 2$ (a hyperbola has two components).

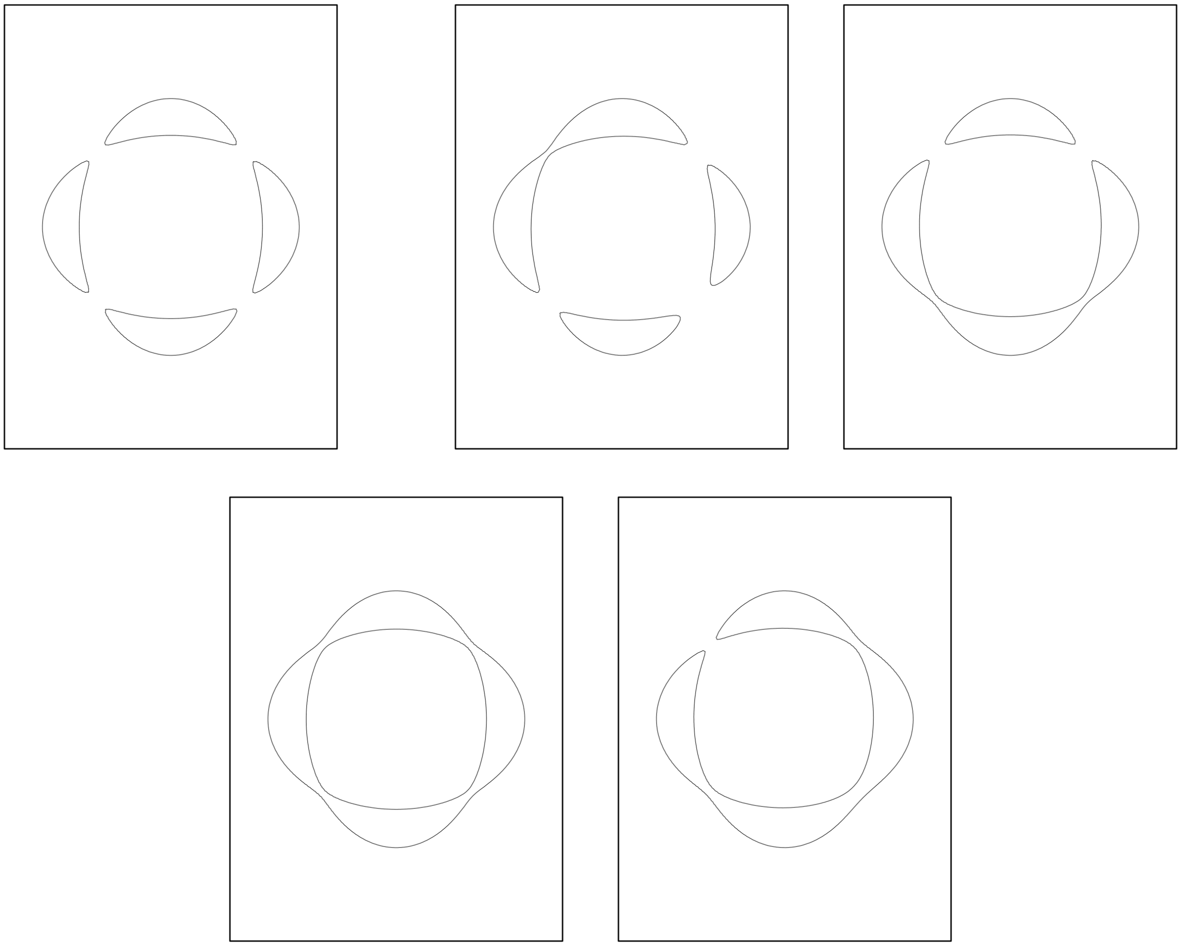


Figure 2.9: One can show that a single curve of degree 4 can achieve all possible real schemes by a series of perturbations of the union of two ellipses with introducing a small additional term into the equation. This figure is taken from [9]

2.4 The Ragsdale Conjecture

Ragsdale ([12]) conjectured inequalities which would restrict the possible arrangements of curves in $\mathbb{RP}^2$, although these would later be disproved by Viro and Itenberg ([7]) as we will later explain in Section 3.6.

Ragsdale considered the separate cases of even and odd degree curves, motivated by a special property: a curve of even degree $m = 2k$ (with $k \in \mathbb{N}$), acts as the common boundary that divides $\mathbb{RP}^2$ in two pieces, i.e., it is two-sided: one part is orientable (denoted $\mathbb{RP}^2_+$) and the other is non-orientable (denoted $\mathbb{RP}^2_-$).

One of the observations Ragsdale made (reproduced from [7]) is the following
Proposition 2.4.1 (Ragsdale observation)**.** *For any Harnack M-curve of even degree $m = 2k$, for $k \in \mathbb{N}$*

$$p = \frac{3k(k-1)}{2} + 1$$

$$n = \frac{(k-1)(k-2)}{2}.$$

These observations motivated Ragsdale's eponymous conjecture:
Conjecture 2.4.2. *For any curve of degree $m = 2k$,*

$$p \leq \frac{3k(k-1)}{2} + 1 \qquad n \leq \frac{3k(k-1)}{2}$$

However, a version of the conjecture for M-curves has neither been proved or disproved as of now:
Conjecture 2.4.3. *For any M-curve of degree $m = 2k$,*

$$p \geq \frac{(k-1)(k-2)}{2} \qquad n \geq \frac{(k-1)(k-2)}{2}$$

Remark 2.4.4. This also offers information on the topology of the sets $\mathbb{RP}^2_+$ and $\mathbb{RP}^2_-$. In particular, the former set has p components, and the latter set $n+1$ components. From this, one may calculate their Euler characteristic of $\mathbb{RP}^2_+$ and $\mathbb{RP}^2_-$.

Chapter 3

The Patchworking Theorem

In this chapter, we introduce the main technique we use for constructing the algebraic surface which will give us the lower bound on the number of maximal number of components. We start by introducing some preliminary concepts, we state Itenberg and Viro's Patchwork theorem, provide its proof and discuss a simple example that illustrates the meaning of the theorem.

3.1 Preliminary Concepts

Definition 3.1.1. *Let $T \subseteq \mathbb{R}^2$ and τ a triangulation of T. We say τ is convex if there exists a continuous function $f : T \to \mathbb{R}$ such that*

(i) $f|_t = f_t$ for some linear function f_t, on each triangle t of T

(ii) f is convex, i.e. $f(x, y) > f_t(x, y)$ for any $(x, y) \notin t$.

Example 3.1.2. Let T be a triangle with vertices $(0, 0)$, $(0, 2)$, and $(2, 0)$, and let τ be the triangulation shown in 3.1 which we will use multiple times going forward. We claim that τ is convex.

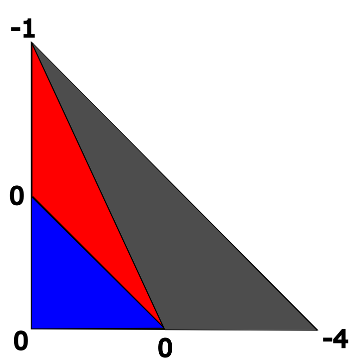


Figure 3.1: A basic convex Triangulation τ that partitions a degree-2 triangle into t_1 (blue), t_2 (red), t_3 (grey)

Proof. Let us label each of the regions of T partitioned by τ as in the Figure 3.1: t_1 (blue), t_2 (red), and t_3 (grey), respectively. Let $f_{t_1}(x,y) = 0$ for all $(x,y) \in t_1$.
Consider the point $(0,2)$. Note that it is not contained within the region t_1, so by construction, $f(0,2) > 0$. Let $f(0,2) = 1$, which implies that $f_{t_2}(x,y) = x + y - 1$. By a similar argument, consider that the point $(2,0)$ is neither in t_1 nor t_2. Hence,

$$\begin{aligned} a + c &= 0 \\ 2b + c &= -1 \\ 2a + c &= -4 \end{aligned}$$

Hence, $a = -4$, $b = \frac{-5}{2}$ and $c = 4$. This gives us

$$f_{t_3} = -4x - \frac{5}{2}y + 4$$

Thus we have shown that $f(x,y) > f_{t_i}, \forall (x,y) \notin t_i$ for $i \in \{1,2,3\}$, i.e. f is convex. $\square$

Going forward, we will also need the notion of homeomorphism. Informally, a homeomorphism is a continuous transformation of one object into another. We omit an extended discussion on topological spaces, but we state the formal definition:
Definition 3.1.3. *Let X, Y be two spaces. A continuous map $f : X \to Y$ is a homeomorphism if it is a bijection and its inverse is also continuous. If there exists such f, we say X and Y are homeomorphic.*

Now, we introduce the notation we will use throughout the chapter. We note that we adopt the same notation introduced in [7] only with slight changes.
Definition 3.1.4. *Let $m \in \mathbb{N}$. Then Δ_m is the triangle with vertices $(0,0)$, $(m,0)$, and $(0,m)$.*
Definition 3.1.5. *Let $m \in \mathbb{N}$. Let T be a triangulation of Δ_m such that all vertices of the triangulation have integer coordinates. Then the map ϵ : Vert(T) $\to \{-1,+1\}$ is the sign distribution of the vertices of the triangulation T of Δ_m, where we denote by Vert(T) the set of vertices of T.*
Definition 3.1.6. *The maps s_x and s_y are the reflection maps across the x-axis, respectively y-axis. The map $s = s_x \circ s_y$ is the reflection across the xy-plane.*
Definition 3.1.7. *Let $m \in \mathbb{N}$. Then $\diamondsuit_m = \triangle_m \cup s_x(\triangle_m) \cup s_y(\triangle_m) \cup s(\triangle_m)$, i.e. the square obtained by joining up the reflections of the triangle Δ_m in all four quadrants.*

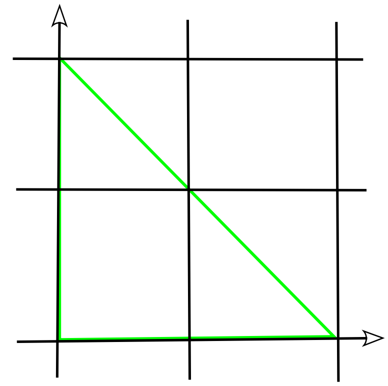
Given a triangulation T of Δ_m (for some fixed $m \in \mathbb{N}$) and a sign distribution ϵ for the vertices of T, we will extend them as follows:

- the triangulation T is extended to a symmetric triangulation of Δ_m;
- the sign distribution ϵ is extended so that when we pass from a vertex to its mirror image then we preserve the sign if and only if the distance is even.

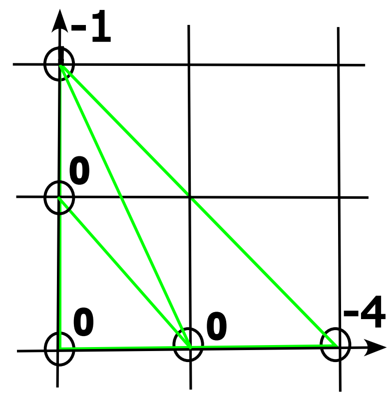
3.2 Statement of the theorem

Before stating the theorem, we will illustrate the process with an example which will hopefully make the statement of the theorem more intuitive.
Example 3.2.1. This example is a more detailed version of the first example in [7].

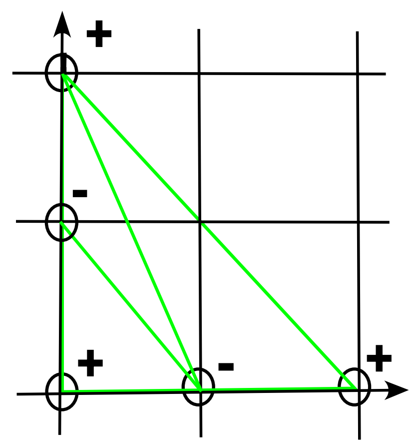
1. We first fix $m \in \mathbb{N}$. Here we take $m = 2$.



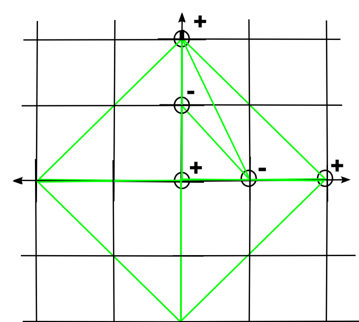
2. We consider the triangle Δ_2 with vertices $(0,0), (2,0), (0,2)$.



3. We now choose a convex triangulation T of Δ_2 such that its vertices have integer coordinates. Recall that this menas there exists a convex piecewise linear function $f \to \mathbb{R}$ which is linear on each triangle small t given by the triangulation T. We choose T shown in the figure. We showed in Example 3.1.2 that this triangulation is indeed convex.

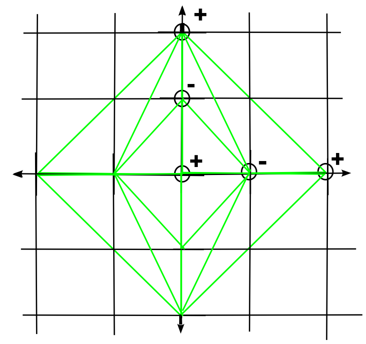


4. We now choose a sign distribution ϵ for $\mathrm{Vert}(T)$. The shown sign distribution is known as the Harnack sign distribution (3.4.2).

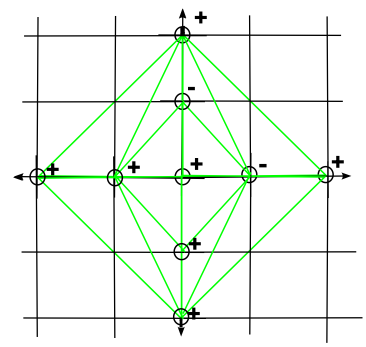


5. We construct the square $\diamondsuit_2$.

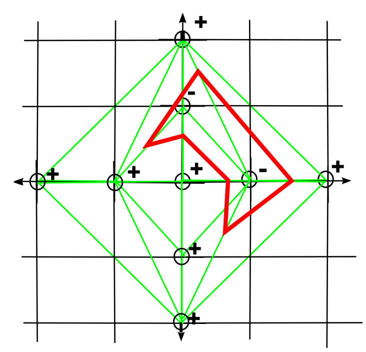
6. We now extend the triangulation T of Δ_2 to a symmetric triangulation of $\diamondsuit_2$.



7. We now extend the sign distribution ϵ by the rule $\epsilon((-1)^a i, (-1)^b j) = (-1)^{a+b}\epsilon(i,j)$ to the vertices of the extended triangulation.



8. Now we analyse each small triangle given by the extended triangulation of $\diamondsuit_2$. If the triangle does not have all the vertices with the same sign, we draw a midsegment (red) which separates the '+'s from the '-'s. The union of all the red segments represents our patchworked curve. The curve has degree $m = 2$.



As explained in the example, in each small triangle given by the extended triangulation of Δ_m, we draw a midsegment which separates the '+' from '-' when this is possible (i.e. when the small triangle's vertices do not all have the same sign). Let us denote by L the union of all these midsegments.

Now, we 'glue' the sides of the square $\diamondsuit_m$ by the reflection map s as shown below: By doing so we obtain a space, let us denote it $\mathcal{P}$, which is homeomorphic to $\mathbb{RP}^2$. Finally, let $\mathcal{L}$ be the image of L in $\mathcal{P}$.

We can now state the patchworking theorem from [7]:
Theorem 3.2.2 (Patchwork Theorem)**.** *Let $m \in \mathbb{N}$. Let T be a convex triangulation of Δ_m. Then there exists a nonsingular real algebraic plane affine curve of degree m and a homeomorphism of the plane $\mathbb{R}^2$ onto the interior of the square $\diamondsuit_m$ which maps the set of real points of this curve onto L. Additionally, there exists a homeomorphism $\mathbb{RP}^2 \to \mathcal{P}$ that maps the set of real points of*

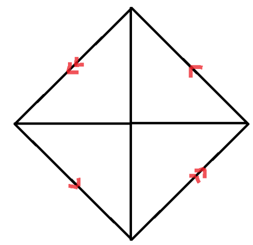


Figure 3.2: Space $\mathcal{P}$

the corresponding projective curve onto $\mathcal{L}$.

Detailed patchworking

We can actually give the explicit form of the polynomials which describe the curve mentioned in the theorem above. We restate the constructions and the main result of the section on T-curves from [7], with slightly changed notation.

Given $m \in \mathbb{N}$, a convex triangulation T together with a function f that witnesses its convexity, and a sign distribution ϵ, we can define, for every $t \in \mathbb{R}$, $t > 0$, the following polynomial:

$$F_t(x, y) = \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i, j) \cdot t^{f(i,j)} \cdot x^i y^j.$$

We also define for every $t \in \mathbb{R}$, $t > 0$ the corresponding homogeneous polynomial:

$$G_t(x_0, x_1, x_2) = x_0^m F_t\left(\frac{x_1}{x_0}, \frac{x_2}{x_0}\right).$$

Theorem 3.2.3 (Detailed Patchwork Theorem)**.** *Let $m, T, f, \epsilon, L,$ and $\mathcal{L}$ be defined as above, and let F_t and G_t be the polynomials constructed above. Then*

1. *F_t defines an affine curve C_t such that the pair $(\mathbb{R}^2, \mathbb{R}C_t)$ is homeomorphic to the pair $(\diamondsuit_m, L)$.*
2. *G_t defines a projective curve C'_t such that the pair $(\mathbb{RP}^2, \mathbb{R}C'_t)$ is homeomorphic to the pair $(\mathcal{P}, \mathcal{L})$.*

3.3 A Proof of The Patchworking Theorem

Let us discuss a sketch of the proof of the patchworking theorem.

Let us remind ourselves the notation we are using (with slight changes). Recall that for some fixed degree, $m \in \mathbb{N}$ and we consider the triangle

$$\Delta_m = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0,\ x + y \leq m\}.$$

Let T be a convex triangulation of Δ_m, i.e. there exists a continuous function $f : \Delta_m \to \mathbb{R}$ such that for every triangle τ of T, there exists a linear function $f_\tau : \mathbb{R}^2 \to \mathbb{R}$ such that

$$f(x,y) = f_\tau(x,y) \quad \text{for } (x,y) \in \tau$$

$$f(x,y) < f_\tau(x,y) \quad \text{for } (x,y) \notin \tau.$$

Let $\mathrm{Vert}(T)$ denote the set of vertices of T, and let $\epsilon : \mathrm{Vert}(T) \to \{\pm 1\}$ be a choice of signs for the vertices of the triangulation T.

We can now start defining the patchworked curve $P(T,\epsilon) \subset \mathbb{RP}^2$. We first define $\diamondsuit_m$ to be the reflection of Δ_m across the coordinate axes (across the x-axis, across the y-axis, and across xy).

We can now extend the triangulation T to $\diamondsuit_m$ by reflection. Similarly, the sign distribution ϵ is extended using the rule

$$\epsilon((-1)^a i, (-1)^b j) = (-1)^{a+b}\epsilon(i,j).$$

We note that we are working in $\mathbb{RP}^2$, so we identify p with $-p$ for all p across the boundary of $\diamondsuit_m$.

Now, we define $P(T,\epsilon)$ so that

$$P(T,\epsilon) \cap \tau = \begin{cases} \varnothing, & \text{if all vertices of } \tau \text{ have the sign } + \\ \varnothing, & \text{if all vertices of } \tau \text{ have the sign } - \\ \text{the midsegment whose endpoints are on edges that have vertices of opposite} \\ \text{signs, otherwise.} \end{cases}$$

Now, for any $t \in \mathbb{R}$, $t > 0$, let us define the real plane curve $C_t = \{F_t(x,y) = 0\}$ where

$$F_t(x,y) = \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j) t^{f(i,j)} x^i y^j.$$

We can now restate the patchworking theorem as follows:
Theorem 3.3.1. *The curve C_t has the same shape as the patchworked curve $P(t,\epsilon)$ for $t > 0$ sufficiently large.*

Proof. It is enough to show this for the first quadrant, $(\mathbb{R}_+)^2$, i.e. we need to show that $C_t \cap (\mathbb{R}_+)^2$ and $P(T,\epsilon) \cap (\mathbb{R}_+)^2$ have the same shape. The result follows by the same argument applied to the other three quadrants.

For every $t \in \mathbb{R}$, $t > 0$ we define the map $\mathrm{Log}_t : \mathbb{R}^2_+ \to \mathbb{R}^2$ by

$$\mathrm{Log}_t(a,b) = (\log_t a, \log_t b).$$

For each $(i,j) \in \mathrm{Vert}(T)$, we define a map $S_{ij} : \mathbb{R}^2 \to \mathbb{R}$ by

$$S_{ij} = f(i,j) + ia + jb.$$

We note that

$$F_t(x,y) = \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j) t^{S_{ij}(\mathrm{Log}_t(x,y))}.$$

This is easy to observe:

$$\begin{aligned} F_t(x,y) &= \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j)\cdot t^{f(i,j)}x^i x^j = \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j)\cdot t^{f(i,j)} t^{i\log_t x} t^{j \log_t y} \\ &= \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j)\cdot t^{f(i,j)+i\log_t x + j\log_t y} = \sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j)\cdot t^{S_{ij}(\mathrm{Log}_t(x,y))}. \end{aligned}$$

We now consider the function $\mathrm{Trop}(F) : \mathbb{R}^2 \to \mathbb{R}$ given by

$$\mathrm{Trop}(F)(a,b) = \max\{S_{ij}(a,b) \mid (i,j) \in \mathrm{Vert}(T)\}.$$

Remark 3.3.2. The function is called a 'tropical polynomial', and it is the Legendre transform of the function f. We omit the exact definitions here, but we refer the reader to [16]. We will expand more on the motivation for using the tropical polynomial in 3.3.7.

For each $(i,j) \in \mathrm{Vert}(T)$, we define $M(i,j) \subset \mathbb{R}^2$ to be the region in which $\mathrm{Trop}(F) = S_{ij}$. The regions $M(i,j)$ are convex polytopes which tile the plane.

We define $Q(T,\epsilon) \subset \mathbb{R}^2$ to be the union of all edges of polytopes such that the regions on either side, $M(i,j)$ and $M(k,l)$, satisfy $\epsilon(i,j) = -\epsilon(k,l)$.

Claim 3.3.3. *The curve $Log_t(C_t \cap \mathbb{R}_+)$ converges to $Q(T,\epsilon)$ as $t \to \infty$. In particular, the two curves have the same shape for $t > 0$ sufficiently large.*

Proof of the claim. If (a,b) lies in the interior of $M(i,j)$, then

$$S_{ij}(a,b) > S_{kl}(a,b)$$

for any $(k,l) \neq (i,j)$. Thus, for large t, we must have that the term $t^{S_{kl}(a,b)}$ is significantly smaller than $t^{S_{ij}(a,b)}$, i.e. for large t the term $t^{S_{kl}(a,b)-S_{ij}(a,b)}$ is significantly smaller than 1.

We now consider the equation $F_t(t^a,t^b) = 0$:

$$\begin{aligned} &F_t(t^a,t^b) = 0 \\ \Leftrightarrow &\sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j)\cdot t^{S_{ij}(\mathrm{Log}_t(t^a,t^b))} = 0 \\ \Leftrightarrow &\sum_{(i,j)\in \mathrm{Vert}(T)} \epsilon(i,j)\cdot t^{S_{ij}(a,b)} = 0. \end{aligned}$$

If we divide by $t^{S_{i,j}(a,b)}$ we obtain

$$\epsilon(i,j) + \sum_{(k,l)\neq(i,j)} \epsilon(k,l) t^{S_{kl}(a,b)-S_{ij}(a,b)} = 0$$

i.e.

$$\epsilon(i,j) + S = 0$$

where S is a finite sum of terms of magnitude significantly smaller than 1. But this is impossible: we can always choose t large enough so that $|S| < |\epsilon(i,j)| = 1$.

So we conclude that $\mathrm{Log}_t(C_t)$ avoids the interior of $M(i,j)$ for all $(i,j) \in \mathrm{Vert}(T)$.

Along the edge between $M(i,j)$ and $M(k,l)$, we have

$$S_{ij} = S_{kl} < S_{pq}$$

for any $(p,q) \neq (i,j),(k,l)$. Dividing the equation $F_t(t^a, t^b) = 0$ by $t^{S_{i,j}(a,b)} = t^{S_{kl}(a,b)}$, we obtain

$$\epsilon(i,j) + \epsilon(k,l) + \sum(S') = 0$$

where S' is a finite sum of terms of magnitude significantly smaller than 1. We can always choose t large enough so that S is much smaller than 1, and so the equality can only be satisfied if $\epsilon(i,j) = -\epsilon(k,l)$. □

Claim 3.3.4. *$Q(T,\epsilon)$ has the same shape as $P(T,\epsilon)$.*

Proof of the claim. We omit the full proof, but we claim that the decomposition of $\mathbb{R}^2$ into the polytopes $M(i,j)$ is dual to the triangulation T of Δ_m, i.e. if one draws a dot in the middle of each $M(i,j)$ and then connects the dots by lines crossing the edges, one obtains T. □

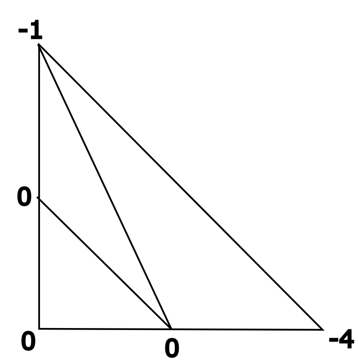
The Patchworking theorem (in the form stated above) now follows from the two claims.

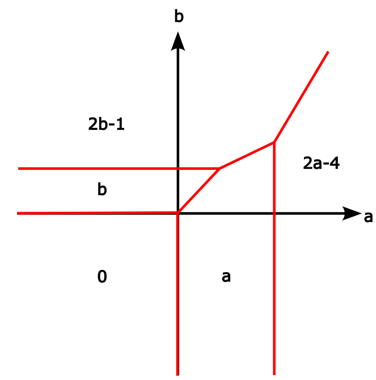
□

Remark 3.3.5. We note that we avoid formalising what we mean by two curves as 'having the same shape'. We also omit the proof of the claim made about duality (3.3.4). We refer the reader to Section 11.5.D of [6] for a complete proof of a slightly more general statement.

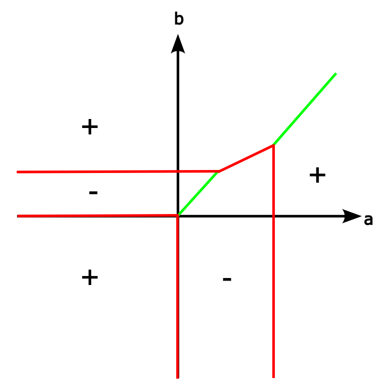
Example 3.3.6. We illustrate some of key concepts introduced in the prior proof with the following example:

Let us consider the triangle with vertices $(0,0)$, $(0,2)$, $(2,0)$ and the triangulation shown on the right. We previously showed (Example 3.1.2) that this triangulation is convex.

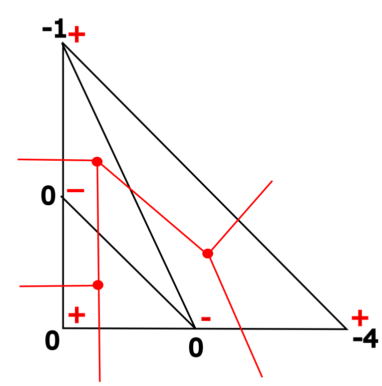




To each vertex (i,j) of the triangulation, we assign a map $S_{ij} = f(i,j) + ia + jb$, as explained in the proof. For example, in oru case we have $S_{00} = 0$, $S_{01} = b$, $S_{02} = 2b - 1$.
We can now construct our tropical polynomial by dividing the plane $\mathbb{R}^2$ into regions such that each region is dominated by some S_{ij}. For example, $S_{20} = 2a - 4 > a = S_{10}$ when $a > 4$. However, $S_{02} = 2b - 1 > 2a - 4 = S_{20}$ when $b > \frac{2a-3}{2}$. In other words, S_{20} dominates S_{10} when $a > 4$, and it is dominated by S_{02} when $b > \frac{2a-3}{2}$.



Recall we assigned a sign $\sigma_{i,j} = \pm$ to each vertex according to our map ϵ. We note that the portions of the tropical polynomial that lie between a positive and negative sign are said to describe the true patchworked curve. We mark these portions in red, and the ones that are omitted in green.



Our tropical polynomial is 'dual' to the patchworked curve.

Remark 3.3.7. We previously claimed that the tropical polynomial has the same shape and is dual to the patchworked curve. Let us sketch an explanation of this claim.

Let us consider the Legendre transform and study how it can be used to ensure duality between our patchworked curve and our tropical curve. Let us define $\text{Trop}(F)$ and the M_{ij}'s as in the proof. Take $M_{i,j} : \{(a,b) : S_{kl} < S_{ij}(a,b)(ab) \quad \forall (k,l) \neq (i,j)\}$. Now we consider the function $P_{ab} : \mathbb{R}^2 \to \mathbb{R}^2$ defined as

$$P_{ab}(z,w) = az + bw.$$

Let $s(a,b) := \max\{s \in \mathbb{R} \mid P_{ab}(z,w) + s \leq -f(z,w), \quad \forall (z,w) \in T\} = -\text{Trop}(F)$

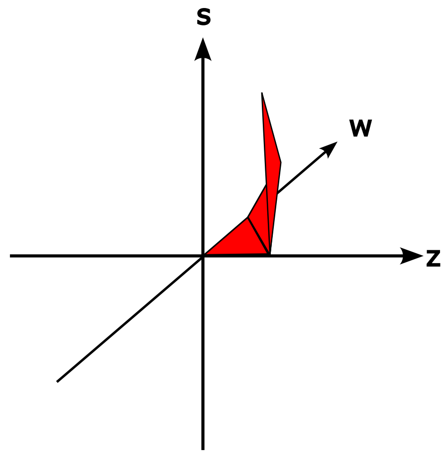


Figure 3.3: The graph of $-f$

Intuitively, we may illustrate the nature of the Legendre transform in the following way. Consider a plane $az + bw + s$. By increasing s, this plane is raised up until it touches our convex triangulation as shown in Figure 3.3. Now recall, by construction, Trop(F) is equivalent to $-s(a,b)$ i.e., the maximum s at which our plane, in effect, crashes into our triangulation.
Geometrically, this collision can occur in 3 ways: the plane may collide at a vertex, along a line segment or with an entire plane.
In the initial case, this would correspond to a vertex of our Tropical curve. One may think of this being the case due to the lack of freedom of movement of the plane should it collide with a triangle of the triangulation. In other words, the plane $az + bw + s$ cannot be rotated meaningfully and still be in contact with the entire plane, as can be done to varying degrees in the latter two cases.
Should collision occur along a line segment, this gives rise to a line segment in our tropical curve, too i.e., the delineating line between two dominating regions.
Finally, the case in which collision occurs at a vertex, gives rise to a region/plane of our tropical curve. In this case, we naturally extend the rationale utilised previously in the more restrictive cases. The plan can be rotated in all directions while still acheiving contact with the vertex.

3.4 Harnack Curves by Patchworking

One application of patchworking is showing that the Harnack Inequality (Theorem 2.2.1) offers a tight upper bound on the number of connected components by constructing such a maximal patchworked curve. Such curves are called Harnack curves, or M-curves. Building on this, we can obtain an upper bound on the number of components of curves in $\mathbb{R}^2$.

We start by formally defining an M-curve:
Definition 3.4.1. *An M-curve is an algebraic curve in $\mathbb{RP}^2$ with a maximal number of components, namely* $\frac{(m-1)(m-2)}{2} + 1$.

We now introduce a particular sign distribution:

Definition 3.4.2. *Let T be a triangulation of Δ_m (for some fixed $m \in \mathbb{N}$). Then the map $\epsilon : Vert(T) \to \{\pm 1\}$ is the Harnack sign distribution if:*

$$\epsilon(i,j) = \begin{cases} \textit{-1}, & \textit{if both } i \textit{ and } j \textit{ are even.} \\ \textit{+1} & \textit{otherwise.} \end{cases}$$

Now, we can restate the theorem from [5] which asserts that such curves always exist:

Theorem 3.4.3. *For any natural number m, there exists a M-curve of degree m.*

Proof. The proof is presented in Itenberg and Viro's paper [7]. We omit it here, but we just remark that Harnack curves can be constructed using patchworking using a special kind of triangulation in which every region is a triangle with area $A = \frac{1}{2}$, and the Harnack sign distribution. ☐

3.5 Bounds on the number of components of degree-2 curves

We are now interested in studying the number of components of curves in $\mathbb{R}^2$ using our new tool, patchworking. We recall that we denote by $C_{2,m}$ the maximum number of components achievable by a degree-m curve in $\mathbb{R}^2$.

Remark 3.5.1. Given a degree-m algebraic plane curve C in $\mathbb{R}^n$, the number of components of C must act as a lower bound on $C_{n,m}$.

We first study to find a lower bound.

Lemma 3.5.2. *We can bound $C_{2,m}$ below by a quadratic function of m.*

Proof. By the remark above, it is enough to find one curve whose number of components is quadratic in m. We will construct such a curve using patchworking.

Let T be the standard triangulation shown in Figure 3.4.

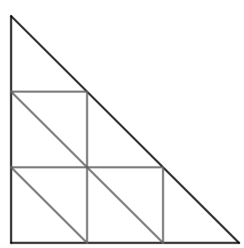


Figure 3.4: Standard triangulation of Δ_3

We can easily prove T is convex. Now, let ϵ be the sign distribution defined as following:

$$\epsilon(i,j) = \begin{cases} +1, & \text{if } i \text{ is odd and } j \text{ is even} \\ -1, & \text{otherwise.} \end{cases}$$

Figure 3.6 shows our patchworked curve inside Δ_6.

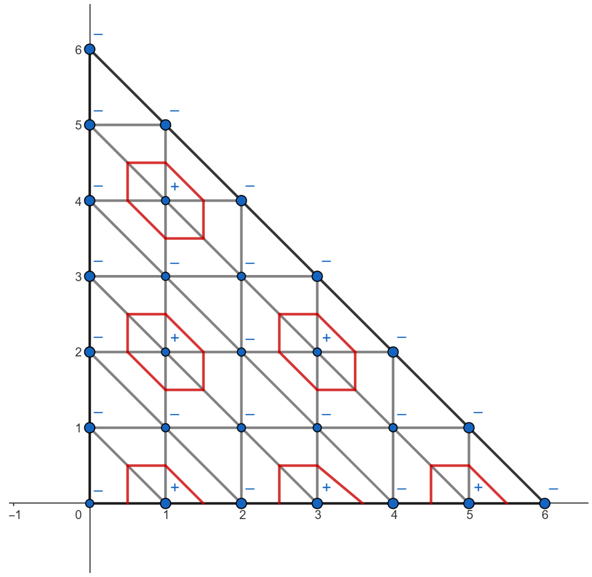


Figure 3.5: The $\mathbb{R}^2_+$ quadrant of our patchworked curve for $m = 6$

Let's count the number of connected components of the patchworked curve completely contained in Δ_m.we can easily observe that the components are arranged as a triangle with vertices $(1, 2)$, $(1, p)$, and $(q, 2)$, where p is the largest even number strictly smaller than m, and q is the largest odd number strictly smaller than m. Our triangle has side length $\lfloor \frac{m-2}{2} \rfloor$, and thus we have

$$\frac{\lfloor \frac{m-2}{2} \rfloor (\lfloor \frac{m-2}{2} \rfloor + 1)}{2}$$

components entirely contained in Δ_m. (We will see later that this is known as the $\lfloor \frac{m-2}{2} \rfloor$-th triangular number).

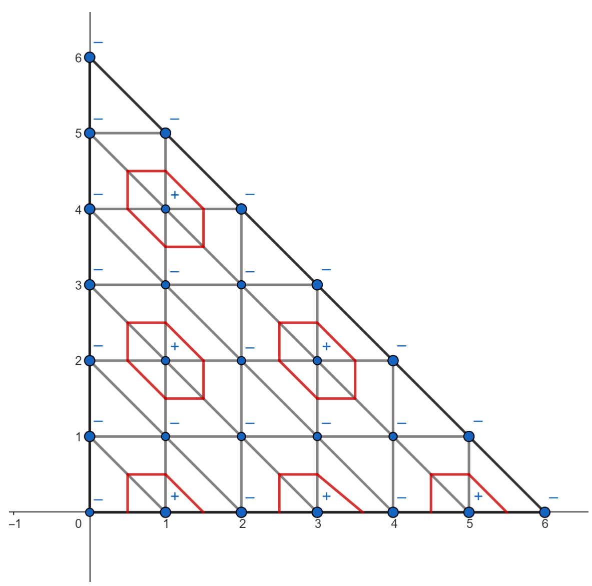


Figure 3.6: The $\mathbb{R}^2_+$ quadrant of our patchworked curve for $m = 6$

Clearly, the expression is quadratic in m, thus we showed our lemma. $\square$

We can now state our first theorem on the bounds of $C_{2,m}$:
Theorem 3.5.3. *$C_{2,m}$ is a quadratic polynomial in m.*

Proof. $C_{2,m}$ is bounded below by a quadratic function in m by the lemma above. From Milnor-Thom (Theorem 2.3.1), we have

$$C_{2,m} \leq m(2m-1),$$

so $C_{2,m}$ is also bounded above by a quadratic polynomial in m. The conclusion follows. □

We now want to study an upper bound on the number of components of curves in $\mathbb{R}^2$.

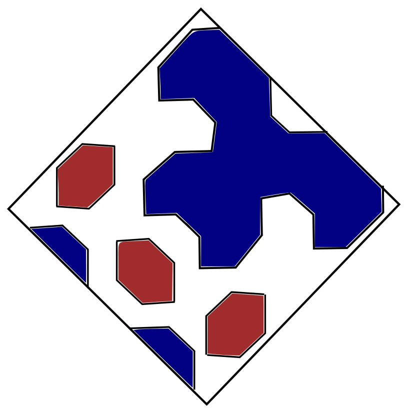


Figure 3.7: A patchworked curve with arbitrary triangulation in $\mathbb{RP}^2$. The blue components are considered to be one connected component in $\mathbb{RP}^2$, but are distinct in $\mathbb{R}^2$

Example 3.5.4. Consider the patchworked curve in $\mathbb{RP}^2$ represented in Figure 3.7. It has 4 components in $\mathbb{RP}^2$, but 6 distinct components in $\mathbb{R}^2$. This is because the components in blue join up at the line at infinity, L_∞.

Now we would now want to improve our upper bound. We claim the following:
Theorem 3.5.5. *Let $m \in \mathbb{N}$. Then*

$$C_{2,m} \leq \frac{(m-1)(m-2)}{2} + m.$$

Proof. Let C be any curve of degree m in $\mathbb{RP}^2$, and let L_∞ be the line at infinity. Let n denote the number of components of the curve C in $\mathbb{RP}^2$, and C_i the i-th component of C.

To obtain the number of components in $\mathbb{R}^2$, we can think of L_∞ as splitting up the components it intersects.

Thus, we have

$$\begin{aligned}|C \cap \mathbb{R}^2| &= |C \setminus (C \cap L_\infty)| \\ &= \sum_{i \in \{1,..,n\}} (|C_i \cap L_\infty) + q_i)\end{aligned}$$

where $q_i = \begin{cases} 1 \text{ if } C_i \cap L_\infty = \varnothing \\ 0 \text{ otherwise.} \end{cases}$

We can now rewrite it as

$$|C \cap \mathbb{R}^2| = |C \cap L_\infty| + |\{C_i \mid C_i \cap L_\infty = \varnothing, i \in \{1, .., n\}\}| \,.$$

Now, we want to find upper bounds on both terms on the right-hand side. We distinguish between two cases:

1. If $C \cap L_\infty = \varnothing$, then:
$$|C \cap L_\infty| = 0$$
and by the Harnack Inequality 2.2.1, we have that
$$|\{C_i \mid C_i \cap L_\infty = \varnothing, i \in \{1, .., n\}\}| \leq \frac{(m-1)(m-2)}{2} + 1.$$
Thus, using equation 3.5, we obtain
$$|C \cap \mathbb{R}^2| \leq \frac{(m-1)(m-2)}{2} + 1$$

2. If $C \cap L_\infty \neq \varnothing$, then by Bézout's theorem we have that
$$|C \cap L_\infty| \leq m.$$
Since C intersects L_∞, at least one of the components C_i's must intersect L, and thus we can slightly adjust the bound given by the Harnack Inequality 2.2.1 by removing this component from the count:
$$|\{C_i \mid C_i \cap L_\infty = \varnothing, i \in \{1, .., n\}\}| \leq \frac{(m-1)(m-2)}{2} + 1 - 1.$$
Thus, we have
$$|C \cap \mathbb{R}^2| \leq \frac{(m-1)(m-2)}{2} + m.$$

Since $m \leq 1$, we have shown that

$$|C \cap \mathbb{R}^2| \leq \frac{(m-1)(m-2)}{2} + m.$$

Since we chose the degree m curve C arbitrarily, we have

$$C_{2,m} \leq |C \cap \mathbb{R}^2| \leq \frac{(m-1)(m-2)}{2} + m.$$

□

Remark 3.5.6. Theorem 3.5.5 gives a significantly tighter bound than Milnor-Thom's theorem

(Theorem 2.3.1). Indeed, for any $m \in \mathbb{N}$, we have that

$$\frac{(m-1)(m-2)}{2} + m \leq m(2m-1)$$

with equality if and only if $m = 1$. The proof is trivial:

$$\frac{(m-1)(m-2)}{2} + m \leq m(2m-1) \Leftrightarrow m^2 - m + 2 \leq 4m^2 - 2m \Leftrightarrow 3m^2 \leq m + 2$$

and if $m > 1$, then

$$3m^2 > 3m = m + 2m > m + 2.$$

3.6 Counterexample of The Ragsdale Conjecture

We now discuss the original application of patchworking, as introduced in [7]. We have previously introduced the Ragsdale conjecture in Section 2.4. Now, we restate the main result showed by I. Itenberg and O.Viro in [7] which disproves the conjecture. We note that the proof of theorem involves constructing curves with the required properties using patchworking.

Theorem 3.6.1. *For each integer $k \geq 1$*

a) there exists a non-singular real algebraic plane projective curve of degree $m = 2k$ with

$$p = \frac{3k(k-1)}{2} + 1 + \left[\frac{(k-3)^2 + 4}{8}\right]$$

b) there exists a non-singular real algebraic plane projective curve of degree $m = 2k$ with

$$n = \frac{3k(k-1)}{2} + \left[\frac{(k-3)^2 + 4}{8}\right]$$

Remark 3.6.2. $\left[\frac{(k-3)^2+4}{8}\right]$ is strictly larger than zero when $k \geq 5$, thus the first counterexample to the conjecture occurs for degree-10 curves.

Proof. The proof can be found in [7]. ☐

Chapter 4

The Three Dimensional Case

In this chapter, we start by presenting a three dimensional version of the patchworking theorem. We then present a choice of triangulation and show it is proper (that is, we will show its convexity which is a much more involved task then in the two dimensional case) and a choice of distributing the signs. Finally, in the context of these choices, we present a way of obtaining a lower bound on the number of components of the patchworked surface. As a short aside, we introduce triangular and tetrahedral numbers and some of their properties which will come in useful when counting components.

4.1 The Three Dimensional Patchworking Theorem

We note that the natural choice for a polyhedron to carry out our patchworking procedure on in three dimensions is a tetrahedron.

Before proceeding to state the theorem, we must first update our notation previously introduced in Section 3.1.

Definition 4.1.1. *Let $m \in \mathbb{N}$.*

1. *Let Δ_m represent the tetrahedron with vertices $(0,0)$, $(0,m)$, $(m,0)$, and (m,m).*
2. *Let s_x, s_y, s_z be the reflection maps. Let $s_{xy} = s_x \circ s_y$, $s_{yz} = s_y \circ s_z$, $s_{xz} = s_z \circ s_z$, and $s_{xyz} = s_x \circ s_y \circ s_z$.*
3. *Let $\diamondsuit_m = \Delta_m \cup s_x(\Delta_m) \cup s_y(\Delta_m) \cup s_z(\Delta_m) \cup s_{xy}(\Delta_m) \cup s_{yz}(\Delta_m) \cup s_{xz}(\Delta_m) \cup s_{xyz}(\Delta_m)$ be the octahedron obtained by joining up the reflections of the tetrahedron in all eight octants.*
4. *Let T be a triangulation of Δ_m. T can be extended to a symmetric triangulation of $\diamondsuit_m$. Let $\epsilon : Vert(T) \to \{\pm 1\}$ be a sign distribution. We can extend ϵ by the rule:*

$$\epsilon((-1)^a i, (-1)^b j, (-1)^c k) = (-1)^{a+b+c}\epsilon(i,j,k).$$

We define L, $\mathcal{P}$, and $\mathcal{L}$ as before (Section 3.1).

We restate the patchworking theorem for the three dimensional case as follows:

Theorem 4.1.2. *Let $m \in \mathbb{N}$. Let T be a convex triangulation of Δ_m. Then there exists a nonsingular real algebraic plane affine surface of degree m and a homeomorphism of the plane*

$\mathbb{R}^3$ *onto the interior of the octahedron* $\diamondsuit_m$ *which maps the set of real points of this surface onto* L. *Additionally, there exists a homeomorphism* $\mathbb{RP}^3 \to \mathcal{P}$ *that maps the set of real points of the corresponding projective curve onto* $\mathcal{L}$.

In other words, fixing some $m \in \mathbb{N}$ and carrying out the equivalaent patchworking procedure produces a patchworked surface of degree m. Let us restate the procedure, this time for theee dimensions:

1. We fix some $m \in \mathbb{N}$. We consider the tetrahedron Δ_m.

2. We choose a convex triangulation T of Δ_m and a sign distribution ϵ of Vert(T).

3. We construct the octahedron $\diamondsuit_m$, and we extend T and ϵ.

4. For every small tetrahedron given by the triangulation T of $\diamondsuit_m$, we consider the plane determined by the midpoints of edges whose vertices have opposite signs. Note that this plane separates the '+'s from the '-'s.

5. Our patchworked surface is composed of these planes.

Remark 4.1.3. We will use T to denote both the triangulation of Δ_m and of $\diamondsuit_m$, but it will be clear from the context which one we are referring to.

4.2 A Higher Dimensional Triangulation

Now, let us consider the tetrahedron (3-simplex) of side length m (Figure 4.1).

$$\Delta_m = \{(x, y, z) \in \mathbb{R}^3 \mid x, y, z \geq 0,\ x + y + z \leq m\}$$

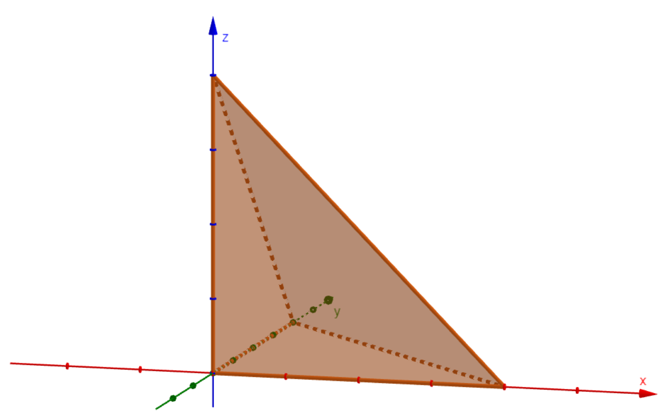


Figure 4.1: The simplex Δ_m

We need to choose a convex triangulation of Δ_m which is a non-trivial task.

We start by choosing a triangulation for the unit cube

$$\mathcal{U} = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1,\ 0 \leq y \leq 1,\ 0 \leq z \leq 1\}$$

as shown in Figure 4.2.

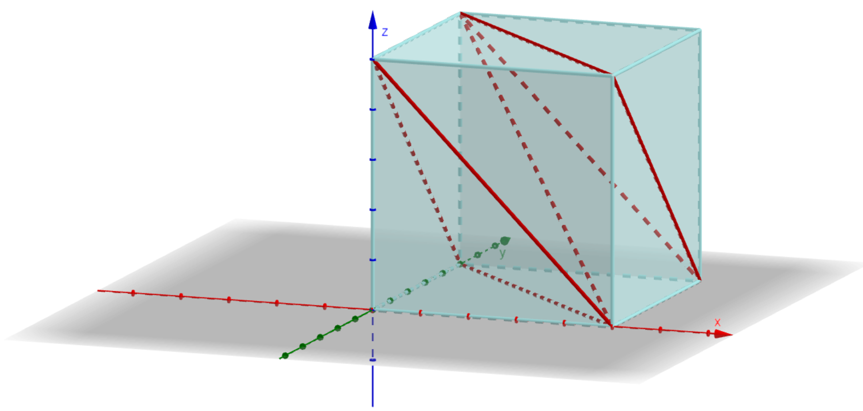


Figure 4.2: The triangulation of the unit cube into 6 tetrahedra

Now, we extend the triangulation of $\mathcal{U}$ to all of $\mathbb{R}^3$ by requiring it to be invariant under reflection in any plane $\{x = k \mid k \in \mathbb{Z}\}$, $\{y = k \mid k \in \mathbb{Z}\}$ or $\{z = k \mid k \in \mathbb{Z}\}$. We will denote this triangulation of $\mathbb{R}^3$ by T.

It is easy to observe that the triangulation T does not 'align' perfectly with our simplex Δ_m. That is, some tetrahedra in T are cut in half by the boundary $x+y+z = m$ of Δ_m. Figure 4.3 illustrates the case of Δ_2.

We will consider only the tetrahedra of T which are entirely contained in Δ_m. As observed above, this will give us only a partial triangulation of Δ_m. Now we would want to show that:

- this partial triangulation of Δ_m is indeed convex;
- we can extend this partial triangulation to a full one which is still convex.

Lemma 4.2.1. *There exists a convex triangulation of Δ_m which includes all tetrahedra given by T which are entirely contained within Δ_m.*

Proof. First, we want to show that T itself is convex. We note that T is obtained by cutting up $\mathbb{R}^3$ along the following planes:

$$\{x = k \mid k \in \mathbb{Z}\},\ \{y = k \mid k \in \mathbb{Z}\},\ \{z = k \mid k \in \mathbb{Z}\}$$

$$\{x + y = 2k + 1 \mid k \in \mathbb{Z}\} \qquad (*)$$

$$\{y + z = 2k + 1 \mid k \in \mathbb{Z}\}$$

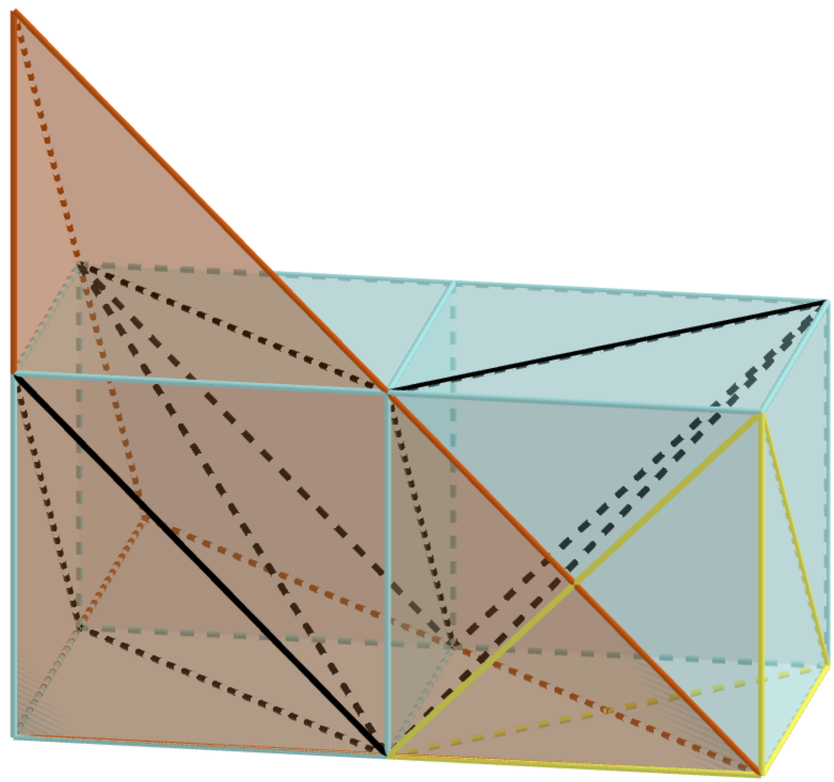


Figure 4.3: The yellow tetrahedron is not entirely contained within Δ_2

and then cutting along the boundary of the octahedron with vertices $(2i+1, 2j, 2k)$, $(2i-1, 2j, 2k)$, $(2i, 2j+1, 2k)$, $(2i, 2j-1, 2k)$, $(2i, 2j, 2k+1)$, and $(2i, 2j, 2k-1)$ which we denote by O_{ijk}, for any $i, j, k \in \mathbb{Z}$.

Consider the function $f : \mathbb{R} \to \mathbb{R}$ with the graph shown in Figure 4.4, i.e. f is given by a line with

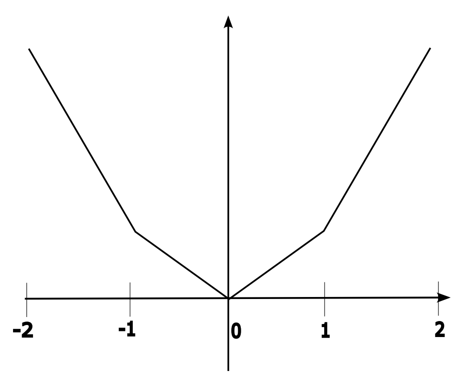


Figure 4.4: The graph of f

slope k on the interval $[0, k]$, for every $k \in \mathbb{N}$. Clearly, f is convex.

Now consider the function $g : \mathbb{R}^3 \to \mathbb{R}$ given by

$$g(x, y, z) = f(x) + f(y) + f(z) + f\left(\frac{x + y - 1}{2}\right) + f\left(\frac{y + z - 1}{2}\right)$$

Note that g is a convex function (since it is a sum of convex functions) and it bounds exactly along the planes $(*)$. Now it remains to make the function bend along the boundaries of the octahedra. For each octahedra O_{ijk}, we define a function $h_{ijk} : \mathbb{R}^3 \to \mathbb{R}$ defined in the following way:

$$\begin{cases} h(x, y, z) = 0, \text{ if } (x, y, z) \notin O_{ijk} \\ h(x, y, z) = \epsilon, \text{ if } (x, y, z) = (2i, 2j, 2k) \text{ (the center of } O_{ijk}) \\ h \text{ is linear on each of the 8 tetrahedron making up } O_{ijk} \end{cases}$$

where ϵ is a small positive constant.

Finally, we define a function $G : \mathbb{R} \to \mathbb{R}$ such that

$$G = g + \sum_{i,j,k\in\mathbb{Z}} h_{ijk}.$$

Then G is convex, and witnesses the convexity of T.

Thus, we have must have that the partial triangulation of Δ_m containing the tetrahedra given by T which are completely contained in Δ_m is convex.

Claim 4.2.2. *We can extend this convex partial triangulation to a full convex triangulation of* Δ_m.

Next, we briefly discuss how this can be done. We note that this fact is used implicitly by Itenberg and Viro in [7].

□

Extending the triangulation near the boundary

We now discuss how we can extend our partial triangulation to a complete triangulation of Δ_m, while preserving convexity. We do not offer a complete proof, but we just sketch the main ideas.

Let $f : \mathbb{Z}^3 \cap \Delta_m \to \mathbb{R}$ be a function. We define $g : \Delta_m \to \mathbb{R}$ to be

$$g(x, y, z) := \max_h h(x, y, z)$$

where the maximum is taken over all maps $h : \Delta_m \to \mathbb{R}$ such that $h(i, j, k) \le f(i, j, k)$ for all $(i, j, k) \in \mathbb{Z}^3 \cap \Delta_m$.

Then g is convex and piecewise-linear. Its domains of linearity are convex polytopes P_α whose vertices lie in $\mathbb{Z}^3 \cap \Delta_m$ such that $\bigcup_\alpha P_\alpha = \Delta_m$.

Lemma 4.2.3. *Suppose* $(i, j, k) \in \mathbb{Z}^3 \cap \Delta_m$. *If we change the function* f *at* (i, j, k) *and nowhere else such that*

$$f(i, j, k) = g(i, j, k) - \epsilon,$$

where $\epsilon > 0$ is sufficiently small, then take the new function g (as defined above, but using the new f), and the new corresponding decomposition $\Delta_m = \bigcup_\alpha P'_\alpha$, then:

- *the polytopes P_α are left unchanged for all α such that $(i, j, k) \notin P_\alpha$;*
- *the polytopes P_α with $(i, j, k) \in P_\alpha$ get subdivided; if P_α is not a simplex, then the subdivision is non-trivial.*

We omit the proof of this lemma, but we make an observation: for $(i', j', k') \in P_\alpha$, the points $\{((i', j', k'), g(i', j', k')) \in \mathbb{R}^4\}$ lie on a hyperplane by definition. If P_α is not a simplex and we perturb the value at (i, j, k), they can no longer lie on a plane, hence the polytope gets nontrivially subdivided.

Corollary 4.2.4. *The subdivision $\Delta_m = \bigcup_\alpha P_\alpha$ can be further subdivided so it is a subdivision into simplices.*

Proof. If some P_α is not a simplex, we perturb the value of f at one of its vertices (as described above) such that it is subdivided. We repeat this process until only simplices remain. ☐

Lemma 4.2.5. *There exists a convex triangulation of Δ_m which coincides with the triangulation T previously described over all simplices contained in Δ_m.*

Proof sketch. Recall that $G : \mathbb{R}^3$ (previously defined) is the function witnessing the convexity of T. We now define $f : \mathbb{Z}^3 \cap \Delta_m \to \mathbb{R}$ as

$$f(i, j, k) := p(i, j, k).$$

We claim that f coincides with T over all simplices contained in Δ_m.

For the rest, there may be some non-simplices. In this case, we can subdivide them further (Corollary 4.2.4) until only simplieces remain, by perturbing f as above. The outcome is convex by construction. ☐

4.3 Choosing a Sign Distribution

Now, we need to choose a sign distribution.

Let $\sigma : \mathrm{Vert}(\mathcal{U}) \to \{\pm 1\}$ be a map which associates a sign to each vertex of the unit cube.

As we extend our triangulation, we also need to extend the sign distribution map to $\sigma : \mathbb{Z}^3 \to \{\pm 1\}$. We impose the requirement that σ is invariant under the same reflections under which the triangulation is invariant, i.e. we have that for any $i, j, k, l \in \mathbb{Z}$

$$\sigma(l - i, j, k) = \sigma(l + i, j, k)$$

$$\sigma(i, l - j, k) = \sigma(i, l + j, k)$$

$$\sigma(i, j, l - k) = \sigma(i, j, l + k)$$

or, equivalently, if we substitute i by $-l + i$, j by $-l + j$, and k by $-l + k$ we have

$$\sigma(2l - i, j, k) = \sigma(i, j, k) = \sigma(i, 2l - j, k) = \sigma(i, j, 2l - k). \tag{4.1}$$

One example of sign distribution with this property is

$$\sigma(i,j,k) = \begin{cases} -1, \text{ if } i,j,k \text{ are all odd} \\ +1, \text{ otherwise.} \end{cases}$$

Cut vertices

For convenience, we define a vertex with a special property:

Definition 4.3.1. *We say a vertex $v = (a,b,c)$ is a cut-vertex if for all vertices $w = (a',b',c')$ adjacent to v (that is, vertices that are connected to v by an edge of the triangulation T) we have that $\sigma(v)\sigma(w) = -1$, i.e. the vertex has a different sign than all its neighbours.*

Remark 4.3.2. Whenever we have a cut-vertex, the patchworked surface will 'cut off' the cut-vertex as pictured in Figure 4.5.

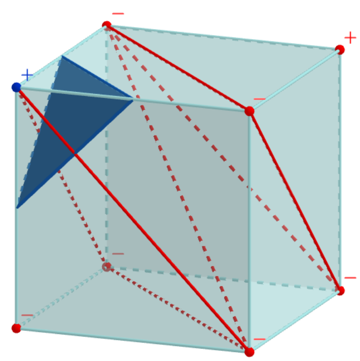


Figure 4.5: Cut vertex and the corresponding section of the patchworked surface (dark blue)

We can now state the following lemma:

Lemma 4.3.3. *Let (a,b,c) be a cut vertex. Then for any $i,j,k \in \mathbb{Z}$ such that*

$$a + 2i \geq 1, \quad b + 2j \geq 1, \quad c + 2k \geq 1$$

$$a + b + c + 2i + 2j + 2k \leq m - 3$$

the patchworked surface has a small spherical component centered at $(a+2i, b+2j, c+2k)$ as shown in Figure 4.6.

Remark 4.3.4. Note that a sphere is 'topologically equivalent' to a sphere, thus we name the components produced by the cut vertices 'spherical components'.

Remark 4.3.5. Note that we have a small sphere for each cut-vertex. Since we want to maximise the number of connected components, we want to find a sign distribution that maximises the total number of cut vertices in the eight octants. We refer the reader to 5.1 for an explanation of why the chosen sign distribution is optimal.

Proof of the lemma. Note that the inequalities simply ensure that all tetrahedra which have the point $(a+2i, b+2j, c+2k)$ as a vertex lie inside the first octant and also inside the region where the triangulaiton agrees with T.

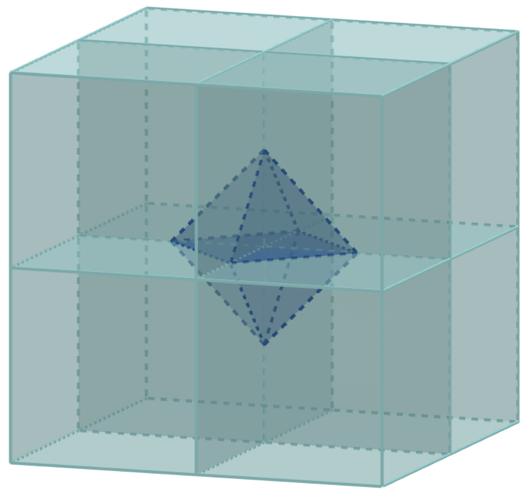


Figure 4.6: A patchworked component centred around a cut-vertex. Note, geometrically this is equivalent to a sphere and shall henceforth be denoted as such.

Since we choose the triangulation and the sign distribution to be invariant under reflection in all planes $\{x = k \mid k \in \mathbb{Z}\}$, $\{y = k \mid k \in \mathbb{Z}\}$, and $\{z = k \mid k \in \mathbb{Z}\}$, then so must be the patchworked surface in the first octant. Using the same argument as for 4.1, we have that the patchworked surface is inviariant under translation by $(2i, 2j, 2k)$. Thus, $(a + 2i, b + 2j, c + 2k)$ must be a cut vertex.

Now, the invariance under reflection in the planes $\{x = a + 2i\}$, $\{y = b + 2j\}$, and $\{z = c + 2k\}$ implies that the patchworked surface includes the spheres described in the statement of the lemma. □

4.4 A Patchworked Example in 3-Dimensions

4.4.1 Triangular and Tetrahedral Numbers

We briefly introduce triangular and tetrahedral numbers, the connection between them and their geometric interpretation.

The triangular number represents the number of dots arranged in an equilateral triangle (Figure 4.7).

Definition 4.4.1 (Triangular numbers)**.** *The n^{th} triangular number, which we denote T_n, represents the the number of unit-radius circles in an equilateral triangle of side-length n.*

It is trivial to show by induction that the n^{th} triangular number T_n is given by

$$T_n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2} \tag{4.2}$$

Similarly, the tetrahedral number represents the number of spheres arranged in a regular tetrahedron (Figure 4.8).

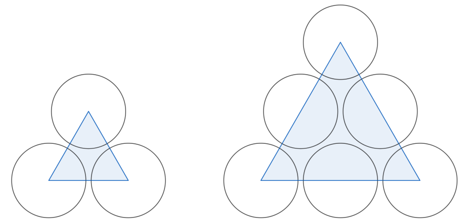


Figure 4.7: The geometric interpretation of T_2 and T_3

Definition 4.4.2 (Tetrahedral numbers)**.** *The n^{th} tetrahedral number, which we denote Te_n, represents the number of unit-radius spheres in a regular tetrahedron of side-length n.*

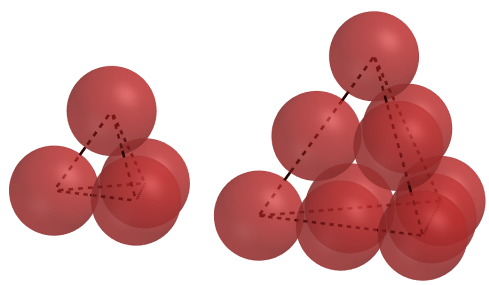


Figure 4.8: The geometric interpretation of Te_2 and Te_3

We can easily observe that the n^{th} tetrahedral number is the sum of the first n triangular numbers:

$$Te_n = \sum_{k=1}^{n} T_k.$$

It is trivial to show by induction that tetrahedral numbers are represented by the formula:
Proposition 4.4.3 (Formula for Tetrahedral numbers)**.** *We can compute the n^{th} tetrahedral number Te_n as*

$$Te_n = \frac{n(n+1)(n+2)}{6} \tag{4.3}$$

4.4.2 Counting Cut Vertices

Now, we are ready to count the number of spherical components of the patchworked surface.

We choose the following sign distribution of the unit cube:

$$\sigma(v) = \begin{cases} -1, & \text{if } v = (0,0,0) \text{ or } v = (1,1,1) \\ +1, & \text{otherwise.} \end{cases}$$

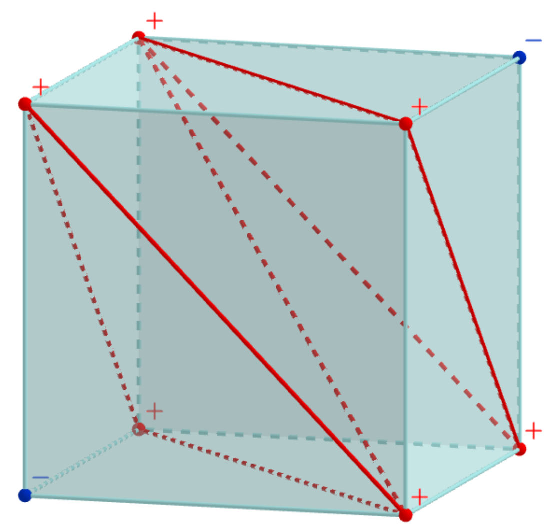


Figure 4.9: Sign distribution and cut vertices of the unit cube in the octant +++

We refer the reader to Appendix 5.1 for the motivation of choosing this particular sign distribution.
Remark 4.4.4. We maintain the same convention for the coordinate axes system (4.2), and going forward we will refer to the specific octants delimited by the axes. For example, by Octant −++ we understand the region $\{(i,j,k) \mid i \leq 0,\ j \leq 0,\ k \geq 0\} \subseteq \mathbb{R}^3$.

Similarly to the two-dimensional case, it is enough to choose signs for the vertices of the +++ octant. The sign distribution in this octant will determine the distributions of all other seven octants.

Thus, for the +++ octant, we can apply the Lemma 4.3.3 directly. For the other octants, the signs must first be reflected accordingly.
Remark 4.4.5. To obtain the sign distribution of some octant X, we just need to reflect the signs across the same planes we would need to reflect the +++ octant to obtain X. For example, to obtain the sign distribution of the unit cube in ++−, we need to reflect the signs across the xy-plane.

We proceed with the argument for the +++ octant . We start by identifying the cut vertices which we mark in dark blue in 4.9.

We observe that a vertex (i,j,k) is a cut vertex if either

$$(1)\begin{cases} i,j,k \text{ are all even} \\ i,j,k > 0 \\ i+j+k \leq m-1 \end{cases} \qquad \text{or} \qquad (2)\begin{cases} i,j,k \text{ are all odd} \\ i,j,k > 0 \\ i+j+k \leq m-1 \end{cases}$$

Remark 4.4.6. We note that the second inequality of both case (1) and case (2) ensure that we are not considering the points on the boundary between octants. This is indeed the intended approach: the behaviour of the surface at the boundary requires understanding the interactions between octants, and we choose to omit it since ultimately we are interested in obtaining a lower bound on the number of components. We can note, however, that there is at least one component at the boundary.

Let $A \subseteq \mathbb{Z}^3$ be the set of points that satisfy case (1) and let N be the largest integer such that

$$2N + 2 + 2 \leq m - 1$$

i.e. $N = \left\lfloor \frac{m-5}{2} \right\rfloor$. Then, the points in A form a tetrahedron with vertices $(2,2,2)$, $(2,2,2N)$, $(2,2N,2)$, and $(2,2,2N)$. Each side of the tetrahedron contains N points in A. Thus the total number of points is given by the N^{th} tetrahedral number, Te_N.

Similarly, let $B \subseteq \mathbb{Z}^3$ be the set of points that satisfy (2), and let M be the largest integer such that

$$2M - 1 + 1 + 1 \leq d - 1$$

i.e. $M = \left\lfloor \frac{m-2}{2} \right\rfloor$. Then, the points in B form a tetrahedron with vertices $(1,1,1)$, $(1,1,2M-1)$, $(1,2M-1,1)$, $(2M-1,1,1)$. Each side of the tetrahedron has M points in B, and so the total number of points is given by the M^{th} tetrahedral number, Te_M. Figure 4.10 illustrates the set B for Δ_7.

So the total number of cut vertices, and thus of spherical components in this octant is

$$Te_N + Te_M = \frac{\left\lfloor \frac{m-5}{2} \right\rfloor (\left\lfloor \frac{m-5}{2} \right\rfloor + 1)(\left\lfloor \frac{m-5}{2} \right\rfloor + 2)}{6} + \frac{\left\lfloor \frac{m-2}{2} \right\rfloor (\left\lfloor \frac{m-2}{2} \right\rfloor + 1)(\left\lfloor \frac{m-2}{2} \right\rfloor + 2)}{6}.$$

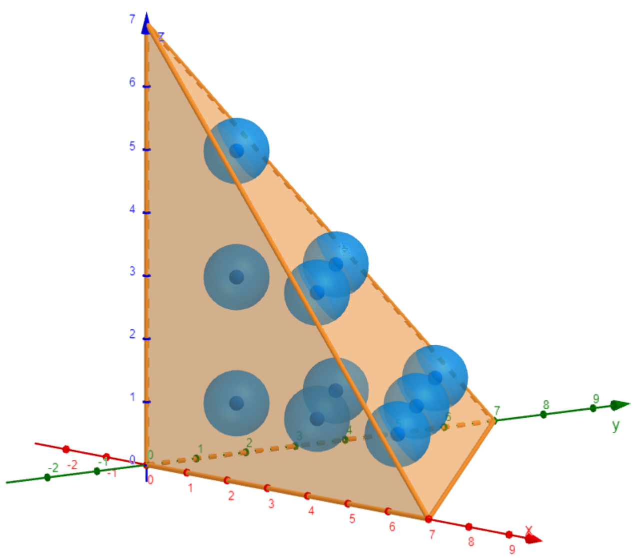


Figure 4.10: The set B for Δ_7. The number of spheres completely contained in Δ_7 is given by the second tetrahedral number, $Te_2 = 4$.

We relate the initial data for the unit cube in the +++ octant 4.9 to each of the remaining octants in the following table.

Octant +\+−	We extend the sign distribution to the region $x > 0, y > 0, z < 0$ (this corresponds to reflection across the xy-plane). We obtain the sign distribution shown in Figure 4.11a for the unit cube.There are no cut-vertices, so this octant does not contribute to the count of spherical components.
Octant +−+	There are no cut vertices in the unit cube, and thus no spherical components see Figure 4.11b. .
Octant −++	There are no cut vertices in the unit cube, and thus no spherical components see Figure 4.11c.
Octant +−−	There are no cut vertices in the unit cube, and thus no spherical components see Figure 4.11d.
Octant −+−	Octant −+− The cut vertices Figure 4.12a of the unit cube are $(0, 0, 0)$ and $(-1, 1, -1)$. We observe that a vertex (i, j, k) is a cut vertex if either $$(1)\begin{cases} i, j, k \text{ are all even} \\ i < 0,\; j > 0,\; k < 0 \\ -i + j - k \leq d - 1 \end{cases} \quad \text{or} \quad (2)\begin{cases} i, j, k \text{ are all odd} \\ i < 0,\; j > 0,\; k < 0 \\ -i + j - k \leq d - 1 \end{cases}$$ By the same argument as in the +++ octant, we have that the number of cut vertices, and thus of spherical components is: $$Te_{\lfloor \frac{m-5}{2} \rfloor} + Te_{\lfloor \frac{m-2}{2} \rfloor} = \frac{\lfloor \frac{m-5}{2} \rfloor (\lfloor \frac{m-5}{2} \rfloor + 1)(\lfloor \frac{m-5}{2} \rfloor + 2)}{6} + \frac{\lfloor \frac{m-2}{2} \rfloor (\lfloor \frac{m-2}{2} \rfloor + 1)(\lfloor \frac{m-2}{2} \rfloor + 2)}{6}.$$
Octant −−+	The cut vertices of the unit cube (Figure 4.12b) are $(0, 0, 0)$ and $(-1, -1, 1)$. We observe that a vertex (i, j, k) is a cut vertex if either $$(1)\begin{cases} i, j, k \text{ are all even} \\ i < 0,\; j < 0,\; k > 0 \\ -i - j + k \leq d - 1 \end{cases} \quad \text{or} \quad (2)\begin{cases} i, j, k \text{ are all odd} \\ i < 0,\; j < 0,\; k > 0 \\ -i - j + k \leq d - 1 \end{cases}$$ Thus, again, the number of spherical components is given by $$Te_{\lfloor \frac{m-5}{2} \rfloor} + Te_{\lfloor \frac{m-2}{2} \rfloor} = \frac{\lfloor \frac{m-5}{2} \rfloor (\lfloor \frac{m-5}{2} \rfloor + 1)(\lfloor \frac{m-5}{2} \rfloor + 2)}{6} + \frac{\lfloor \frac{m-2}{2} \rfloor (\lfloor \frac{m-2}{2} \rfloor + 1)(\lfloor \frac{m-2}{2} \rfloor + 2)}{6}.$$
Octant −−−	There are no cut vertices in the unit cube, and thus no spherical components see Figure 4.12c.

Remark 4.4.7. Note that in order to construct the patchworked surface, we would need to reflect the triangulation of the unit cube in each octant (in addition to reflecting the sign distribution). However, since we are interested only in counting cut vertices, this step is not necessary since it does not change the count.

Let us denote the total number of spherical components centred around cut vertices which belong to exactly one octant by S_m. Considering the analysis above, we have that

$$S_m = 3\left(Te_{\lfloor \frac{m-5}{2} \rfloor} + Te_{\lfloor \frac{m-2}{2} \rfloor}\right) = \frac{\lfloor \frac{m-5}{2} \rfloor (\lfloor \frac{m-5}{2} \rfloor + 1)(\lfloor \frac{m-5}{2} \rfloor + 2) + \lfloor \frac{m-2}{2} \rfloor (\lfloor \frac{m-2}{2} \rfloor + 1)(\lfloor \frac{m-2}{2} \rfloor + 2)}{2}.$$

Finally, we distinguish between two cases.

1. For even degree curves, i.e. $m = 2k$, for $k \in \mathbb{N}$ we have that $\lfloor \frac{m-5}{2} \rfloor = \frac{d-6}{2}$ and $\lfloor \frac{m-5}{2} \rfloor = \frac{m-2}{2}$, and thus we can simplify the previous expression for S_m:
$$S_m = \frac{1}{8}(m^3 - 6m^2 + 20m - 24). \tag{4.4}$$

2. For odd degree curves, i.e. $m = 2k + 1$, for $k \in \mathbb{N}$, we have that $\lfloor \frac{m-5}{2} \rfloor = \frac{m-5}{2}$ and $\lfloor \frac{m-2}{2} \rfloor = \frac{d-3}{2}$, and thus
$$S_m = \frac{1}{8}(m^3 - 6m^2 + 11m - 6). \tag{4.5}$$

Therefore, we obtained a lower bound on the maximal number of connected components. As explained above (Remark 4.4.6), we know that there is at least one additional component at the boundary between octants.

Remark 4.4.8. After the analysis of the octants, we are able to say more about the boundary of our patchworked surface: we now know that the number of components is at least one and at most km^2 for some constant k, i.e. quadratic in m. Since the lower bound we obtained is a cubic in m, including the boundary components would not improve the bound tremendously.

We can now state the main theorem of our paper.

Theorem 4.4.9. *Let $m \in \mathbb{N}$. Then*

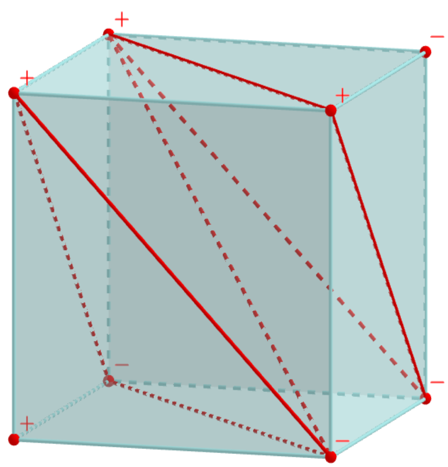
1. *For $m = 2k$ with $k \in \mathbb{N}$,*
$$C_{3,m} \geq \frac{1}{8}(m^3 - 6m^2 + 20m - 24) + 1$$

2. *For $m = 2k + 1$ with $k \in \mathbb{N}$,*
$$C_{3,m} \geq \frac{1}{8}(m^3 - 6m^2 + 11m - 6) + 1.$$

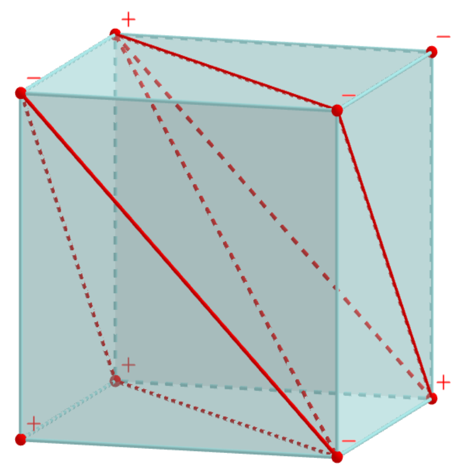
Remark 4.4.10. We note we can bound $C_{3,m}$ below using Theorem 4.4.9 and above using Milnor-Thom's Theorem (2.3.1) which tell us that

$$C_{3,m} \leq m(2m-1)^2.$$

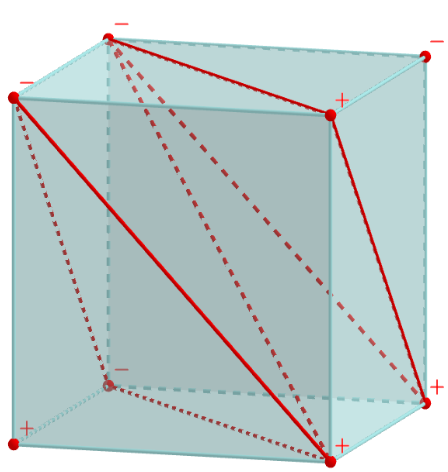
So, $C_{3,m}$ is bounded by two cubics, and thus it must be cubic.



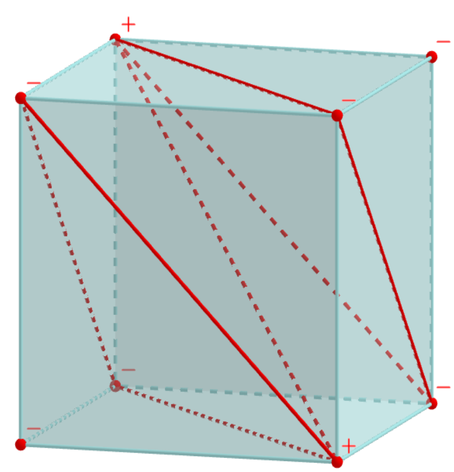
(a) +\+−



(b) +−+

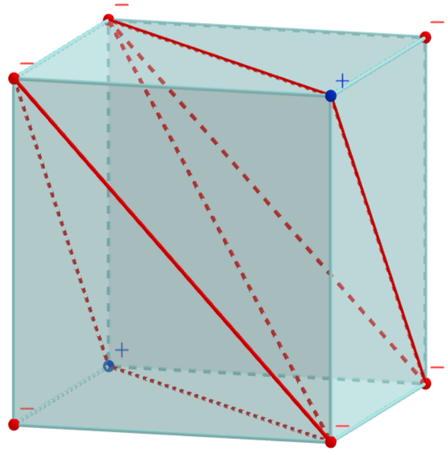


(c) −++

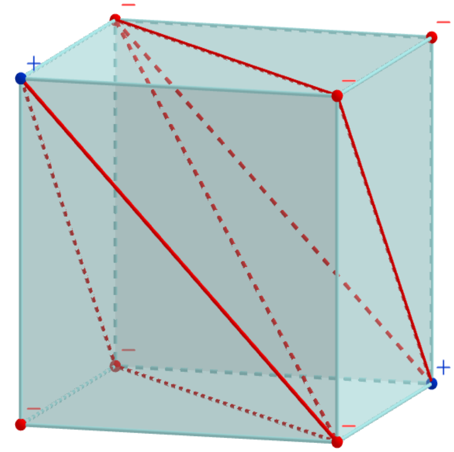


(d) +−−

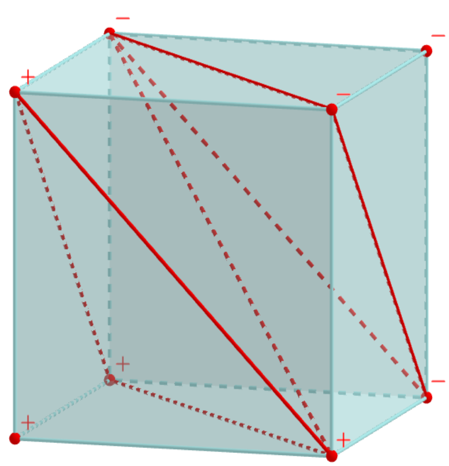
Figure 4.11: The sign distribution and cut vertices of the unit cube in different octants



(a) −+−



(b) −−+



(c) −−−

Figure 4.12: The sign distribution and cut vertices of the unit cube in different octants

Chapter 5

Appendix

5.1 Finding the optimal sign distribution

We present a Python script that we implemented in order to find an optimal sign distribution of the unit cube, given the triangulation shown in 4.2.

The main functionality is:

- counting the number of cut vertices for a specific sign distribution of the unit cube given as user input (the signs of the vertices are input as -1 or 1 in the order shown in Figure 5.1)
- finding the sign distribution which maximises the total number of cut vertices in all eight octants

Since the search of a sign distribution is exhaustive, we can ensure that we choose an optimal one for our analysis.

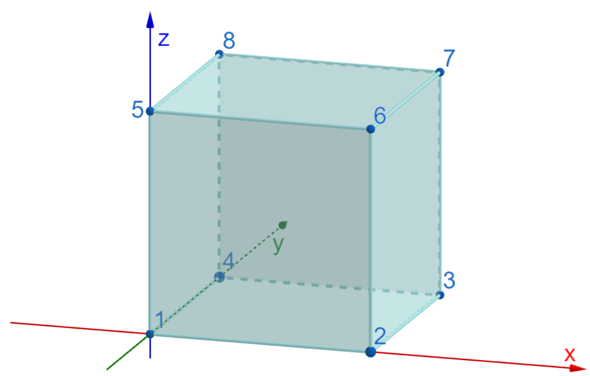


Figure 5.1: Order of the vertices of the unit cube

Remark 5.1.1. We note that the optimal sign distribution is not unique.

```
import numpy as np
import itertools

def count_cut_vertices_current_octant(new_signs):

    isCutVertex = [True] * 8

    for i in range(8):
        for j in range(i + 1, 8):
            # if the vertices i and j are connected and they have the same sign,
            # then neither of them is a cut vertex
```

```
            if edges[i][j] and new_signs[i] == new_signs[j]:
                isCutVertex[i] = False
                isCutVertex[j] = False

    # return the number of cut vertices in the current octant
    return sum(isCutVertex)

def count_total_nr_cut_vertices(signs, per_octant=False):

    # Define reflections across xy, yz, and xz planes
    xyrefl = np.array([-1, -1, -1, -1, 1, 1, 1, 1])
    yzrefl = np.array([-1, 1, 1, -1, -1, 1, 1, -1])
    xzrefl = np.array([-1, -1, 1, 1, -1, -1, 1, 1])

    reflections = [yzrefl, xzrefl, xyrefl]

    total_cut_vertices = 0

    if per_octant:
        print("Number of cut vertices in")
    for reflacrossxy in [True, False]:
        for reflacrossyz in [True, False]:
            for reflacrossxz in [True, False]:
                octant=''
                new_signs = np.copy(signs)
                for (i, reflacross) in enumerate([reflacrossyz, \
                    reflacrossxz, reflacrossxy]):
                    if reflacross:
                        new_signs = np.multiply(new_signs, reflections[i])
                        octant += '-'
                    else:
                        octant += '+'
                nr_cut_vertices = count_cut_vertices_current_octant(new_signs)

                if per_octant:
                    print ("Octant", octant, ": ", nr_cut_vertices)

                total_cut_vertices += nr_cut_vertices

    return total_cut_vertices

# Find a sign distribution that produces the maximum number of cut vertices
def find_max_sign_distr():
    max_cut_vertices = 0
    max_sign_distr = []
    # Generate all possible sign distributions
    for vertex_signs in itertools.product([-1, 1], repeat=8):
```

```
        sign_distr = np.asarray(vertex_signs)
        total_cut_vertices = count_total_nr_cut_vertices(sign_distr)
        if (total_cut_vertices > max_cut_vertices):
            max_cut_vertices = total_cut_vertices
            max_sign_distr = [sign_distr]
        elif (total_cut_vertices == max_cut_vertices):
            max_sign_distr.append(sign_distr)

    print ("Maximum number of cut vertices is ", max_cut_vertices)
    print ("This can be achieved with the following sign distribution(s)")
    for i in range(len(max_sign_distr)):
        print(max_sign_distr[i])


# Choice of triangulation

# Mark std_triang[i][j] with True iff the triangulation
# of the unit cube connects vertices i and j

# standard triangulation
std_triang = [[True, True, False, True, True, False, False, False],
         [True, True, True, True, True, True, False, True],
         [False, True, True, True, False, True, True, True],
         [True, True, True, True, True, False, False, True],
         [True, True, False, True, True, True, False, True],
         [False, True, True, False, True, True, True, True],
         [False, False, True, False, False, True, True, True],
         [False, True, True, True, True, True, True, True]]

edges = std_triang

# Choice of signs for the unit cube
user_signs = np.zeros(8)
for i in range(8):
    sign = int(input())
    user_signs[i] = sign

nr_cut_vertices = count_total_nr_cut_vertices(user_signs, per_octant=True)
print("The total number of cut vertices for the \
     chosen sign distribution: ", nr_cut_vertices)

find_max_sign_distr()
```

5.2 Figures

We present here a list of interactive 3d-figures for the more complex figures included in the text which were created using the Geogebra software.

1. Example of a cut vertex
2. Spherical component centred around a cut vertex belonging to four unit cubes

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